

FRED Economic Monitor

Federal Reserve Bank of St. Louis Data

Report generated: May 17, 2026

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92 series loaded

EXECUTIVE SUMMARY

A synthesis of current economic conditions across six dimensions. Individual readings matter less than combinations and trends. Color indicates risk status: green = normal, yellow = elevated, red = stressed. Use the bookmarks panel in your PDF viewer to navigate between sections.

OVERALL ASSESSMENT: HIGH STRESS

Core inflation (3.0% YoY) remains above target, limiting policy flexibility. The Fed cannot ease without risking re-acceleration.

Labor market solid: unemployment at 4.3%, payrolls adding +115K/month. No demand-destruction signal in employment data.

Yield curve positively sloped (0.5%) with low recession probability (1.8%%). Historical leading indicators not signaling near-term contraction.

Credit markets functional: HY spreads 2.8%, IG spreads 0.8%. Refinancing conditions normal — no systemic funding stress.

CRE lending standards easing or stable (-3.3% net tightening). Banks not restricting commercial real estate credit — no forward stress signal.

Equity valuations at historically stressed levels (CAPE 41.7). Among the highest readings of the past century — historically followed by decade-long below-average real returns. CAPE does not predict timing.

Policy flexibility severely constrained: debt service 33.0% of receipts; inflation expectations 2.5%. Fiscal and monetary stabilization tools are simultaneously impaired.

Historical Comparison — Current vs. 1 Year Ago vs. Pre-COVID

Key indicators compared across three time points. Current cell shading reflects the current risk status (green / yellow / red). Core CPI shown as year-over-year percent change; all others as reported level values.

Indicator	Current	1 Year Ago	Pre-COVID (Dec 2019)
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Core CPI	2.7%	2.8%	2.3%
Unemployment Rate	4.3%	4.3%	3.6%
10Y Treasury Yield	4.5%	4.4%	1.8%
HY Credit Spread	2.8%	3.2%	—
Fed Funds Rate	3.6%	4.3%	1.6%
S&P; 500 CAPE	41.7	35.1	30.3
Federal Debt / GDP	122.6%	118.8%	105.8%
VIX	17.3	17.2	12.6
CRE Lending Stds (net %)	-3.3%	10.9%	10.7%
10Y Inflation Expectation	2.5%	2.3%	1.6%

This summary is generated from threshold-based rules applied to current data. It is descriptive, not predictive. Readings that appear benign in isolation may be significant in combination — see [Crisis Watch](#) and [System Resilience](#) sections for multi-indicator composite analysis.

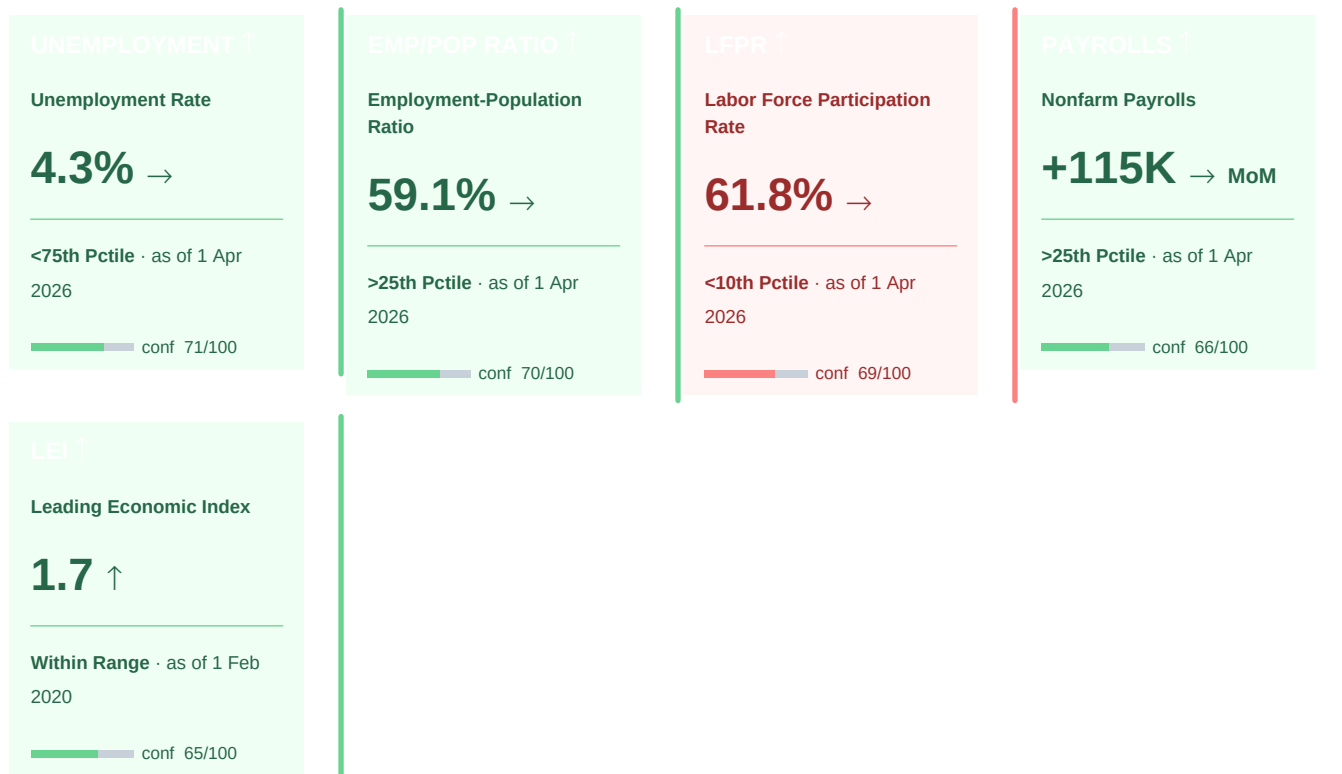
SUMMARY — ALL INDICATORS

Each card shows the current reading, scoring status, and confidence score. Click the indicator name to jump to the detailed section. Labels reflect the scoring method: percentile-scored indicators show pctile bands (<75th / 75–90th / >90th); z-score indicators show σ -distance; fixed-threshold indicators show Within Range / Elevated / Stressed. Context and combinations matter more than any single reading. See the Methodology Appendix for full scoring details.

INFLATION & PRICES



LABOR MARKET



MARKET VALUATION



MARKETS



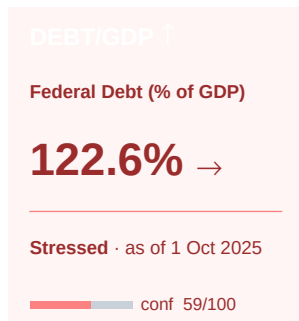
INTEREST RATES



FINANCIAL STRESS



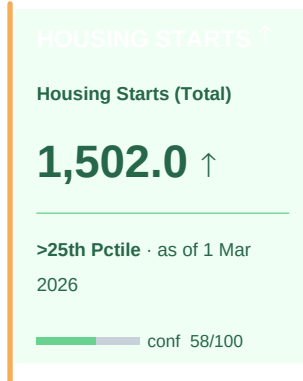
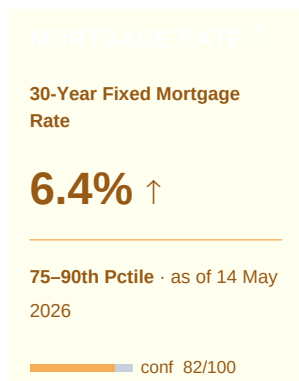
FISCAL HEALTH



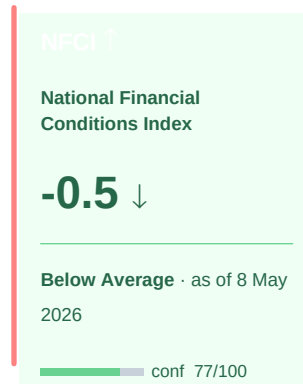
MONEY SUPPLY



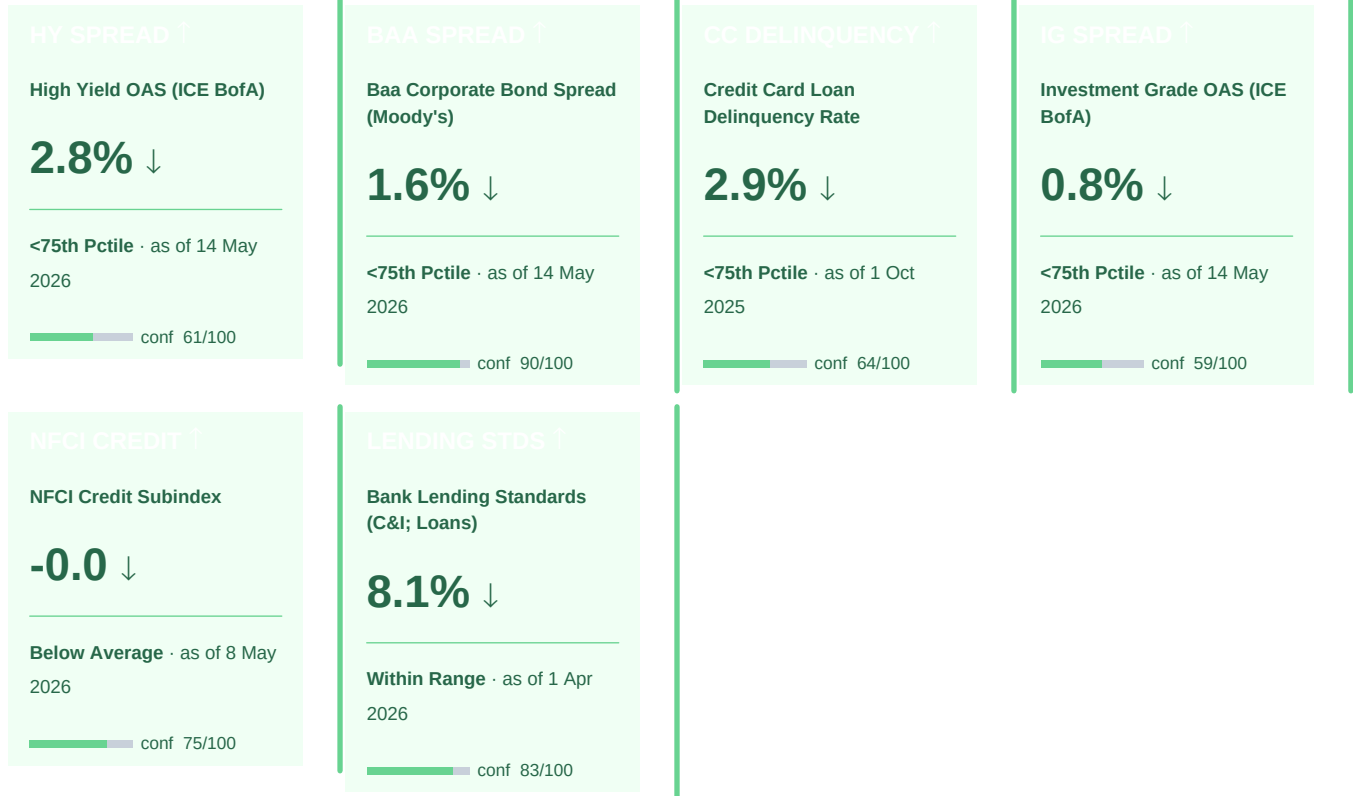
HOUSING MARKET



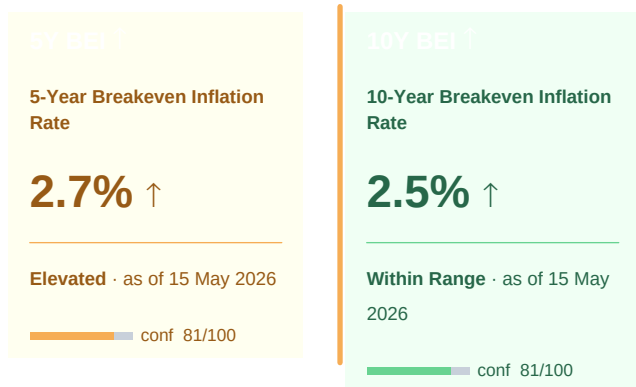
LIQUIDITY & FUNDING STRESS



CREDIT MARKET FUNCTIONALITY



POLICY CONSTRAINTS



CRISIS WATCH

This section synthesizes multiple indicators into six structural dimensions. Color coding is descriptive, not predictive — elevated readings indicate conditions that have historically preceded stress, not guaranteed outcomes.

3 dimension(s) stressed, 1 elevated, 3 within normal range.

TREASURY STRESS

Stressed

10Y Yield **4.5%** 14 May 2026

Yield Curve **0.5%** 15 May 2026

CRE Lending
Std's **-3.3%** 1 Apr 2026

RECESSION RISK

Within Range

NY Fed Rec. **1.8%** 1 Mar 2026
Prob.

Yield Curve **0.5%** 15 May 2026

INFLATION PERSISTENCE

Elevated

Core CPI YoY **3.0%** 1 Apr 2026

Headline CPI **3.9%** 1 Apr 2026
YoY

LABOR WEAKNESS

Within Range

Unemployment **4.3%** 1 Apr 2026
Rate

Payrolls MoM **+115K** 1 Apr 2026

Leading Econ **1.7** 1 Feb 2020
Index

FINANCIAL STRESS

Within Range

VIX **17.3** 14 May 2026

CRE Lending **-3.3%** 1 Apr 2026
Std's

DEBT SUSTAINABILITY

Stressed

Federal Debt / **122.6%** 1 Oct 2025
GDP

10Y Yield **4.5%** 14 May 2026

MARKET VALUATION

Stressed

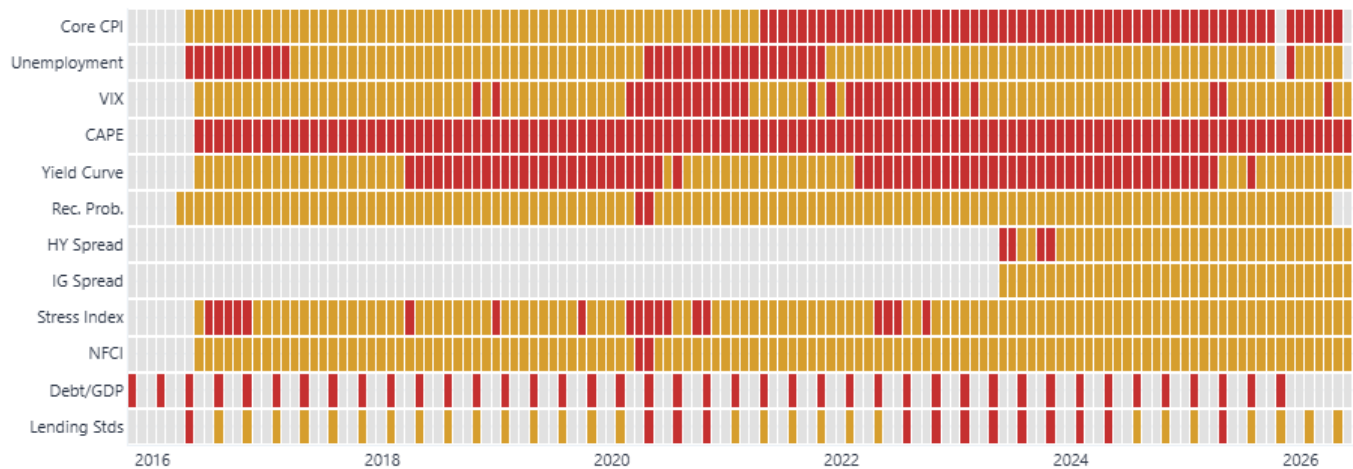
CAPE (Shiller **41.7** 1 May 2026
P/E)

Trailing P/E **31.9** 1 May 2026

Historical Indicator Risk Status

The heatmap below shows the risk classification of twelve key indicators over the past ten years (monthly). Green = normal; yellow = elevated; red = stressed. Clusters of color indicate correlated deterioration across multiple indicators.

Indicator Risk Status — 10-Year History (Monthly)

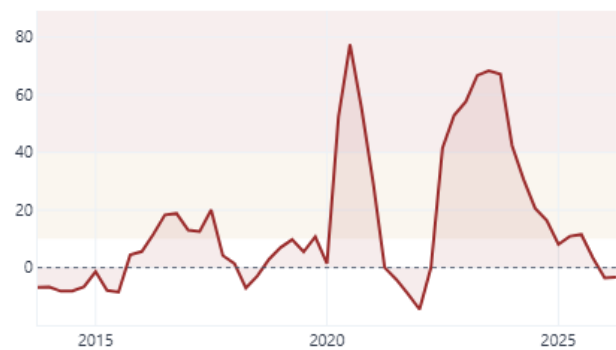


Indicator Detail

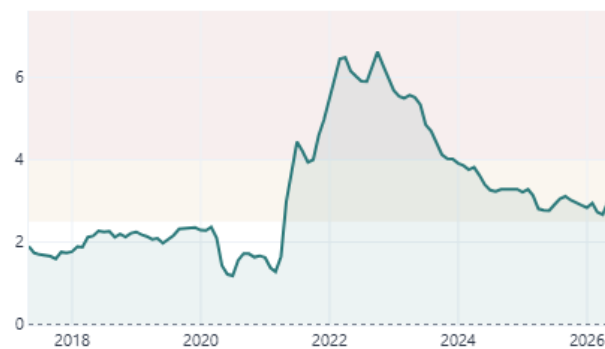
10-Year Treasury Yield %



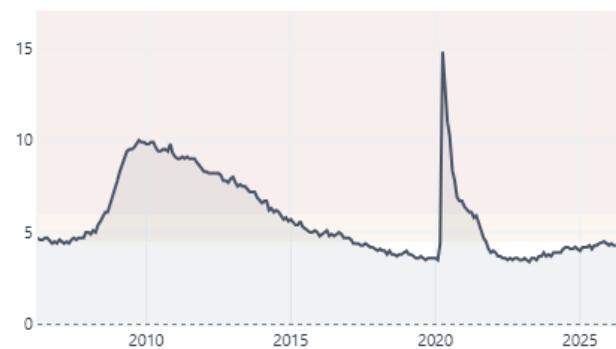
CRE Lending Standards — Net % Tightening



Core CPI YoY %



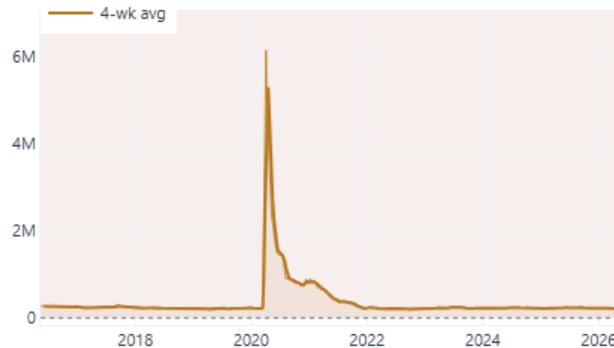
Unemployment Rate %



Conference Board Leading Economic Index (MoM %)



Initial Jobless Claims — 4-Week MA



Structural Deterioration vs. Acute Crisis

Structural deterioration develops over years: rising debt loads, persistent above-target inflation, labor market erosion, or deteriorating credit quality in key sectors. These imbalances accumulate quietly, often masked by growth momentum or asset price appreciation, until they constrain policy options.

The 2008 Global Financial Crisis illustrates the distinction: CRE and residential delinquencies climbed for 18+ months before Lehman's collapse compressed the acute phase into a single weekend. The structural fragility had been building; the trigger was catalytic, not causal.

Historically Dangerous Indicator Combinations

Debt-Rate Spiral

High government debt/GDP + rapidly rising interest rates → expanding debt service costs → crowding out productive investment. The critical metric is debt service as a share of revenue, not the absolute debt level.

Credit Stress Cascade

Rising loan delinquencies + elevated market volatility + tightening lending standards → credit contraction → demand compression → self-reinforcing slowdown.

Stagflation Trap

Persistent above-target inflation + softening labor market → contradictory signals for central banks → elevated policy error risk. The 1970s remain the benchmark episode: dual mandates in conflict with no clean resolution available.

Confidence Breakdown

When multiple dimensions show stress simultaneously, policy space narrows. Fiscal tools are constrained by debt levels. Monetary tools are constrained by inflation. Each instrument has fewer degrees of freedom.

SYSTEM RESILIENCE & POLICY DEPENDENCY

The modern financial system is increasingly dependent on liquidity provision, refinancing capacity, and policy intervention. Structural fragility often emerges not from isolated economic weakness, but from deterioration in the system's ability to absorb shocks without extraordinary stabilization measures.

System Resilience Dimensions

LIQUIDITY & FUNDING	CREDIT FUNCTIONALITY	POLICY FLEXIBILITY
Stressed	Within Range	Stressed
Financial Stress Index -0.8 8 May 2026	High Yield Spread 2.8% 14 May 2026	5-Yr Inflation Expectations 2.7% 15 May 2026
Nat'l Financial Conditions -0.5 8 May 2026	Investment Grade Spread 0.8% 14 May 2026	10-Yr Inflation Expectations 2.5% 15 May 2026
CP-Treasury Spread 0.03% 13 May 2026	C&I; Loan Standards 8.1% 1 Apr 2026	Interest / Govt Receipts 33.0% 1 Oct 2025
Fed Reverse Repo (\$B) \$1B 15 May 2026	CRE Loan Standards -3.3% 1 Apr 2026	Real Fed Funds Rate 0.7% 1 Apr 2026
	Financial Conditions: Credit -0.0 8 May 2026	

System Absorption Capacity Summary

CATEGORY	STATUS	KEY METRIC	CURRENT READING
Liquidity	Within Range	Financial Stress / Nat'l Financial Conditions	-0.8 / -0.5
Funding Stability	Stressed	CP-Treasury Spread / Reverse Repo	0.03% / \$1B
Policy Flexibility	Stressed	Interest / Govt Receipts	33.0%
Credit Functionality	Within Range	High Yield Spread / CRE Loan Standards	2.8% / -3.3%
Inflation Anchoring	Elevated	5Y / 10Y Breakeven Inflation	2.7% / 2.5%

Refinancing & Liquidity Risk Assessment

REFINANCING & LIQUIDITY RISK — ELEVATED · 2 stress condition(s) triggered. Persistent combinations — not single spikes — are the primary concern.

- Fed reverse repo near depletion and declining — note: RRP drawdown can reflect reserve redistribution or MMF allocation shifts, not necessarily system-wide liquidity reduction
- Federal debt service at 33.0% of receipts — fiscal stabilization capacity materially constrained

Note: This assessment emphasizes persistence and combinations. A single elevated indicator is less significant than two or more deteriorating in tandem. Temporary spikes resolve quickly; structural regime shifts do not.

Structural Stress vs System Absorption Capacity

Structural Fragility — conditions that accumulate slowly and reduce the system's margin of safety: elevated debt loads, persistent inflation, labor market deterioration, demographic stagnation, and tightening credit standards. These develop over years and are often masked by asset appreciation or credit availability.

Absorption Capacity — the system's active ability to withstand shocks without requiring progressively larger interventions: functioning credit markets, liquidity provision, policy flexibility, anchored inflation expectations, and stable funding markets. When this deteriorates, structural imbalances become acute crises rather than manageable headwinds.

Modern markets can tolerate elevated structural imbalances for extended periods if refinancing channels remain functional and policy credibility remains intact. Acute crises emerge when structural fragility converges with deterioration in liquidity, credit, or stabilization capacity — not from structural imbalances alone.

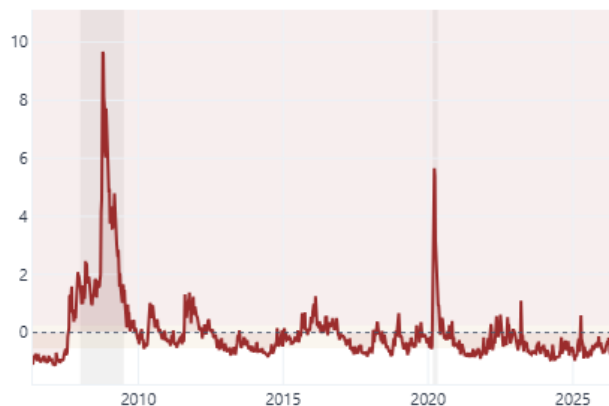
LIQUIDITY & FUNDING STRESS

This section monitors the financial plumbing that supports market functioning: short-term funding markets, interbank credit conditions, and the availability of safe-haven liquidity. Stress here typically appears before it surfaces in equity prices, unemployment, or GDP data — making these indicators leading signals of systemic deterioration.

Financial Stress Indices

The St. Louis Fed FSI and Chicago Fed NFCI both aggregate dozens of financial market variables into a single stress reading. Both are indexed around 0 (average conditions). Positive readings indicate tighter-than-normal conditions; readings above 1.0-1.5 have historically coincided with significant market dislocations.

St. Louis Fed Financial Stress Index (current: 10th percentile)



Chicago Fed National Financial Conditions Index (current: 36th percentile)

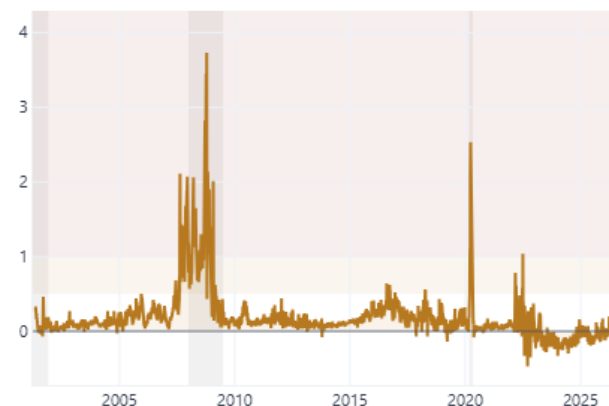


Shaded vertical bands mark NBER-defined recession periods. Colored horizontal bands show the 25th/75th historical percentile ranges — providing context for whether current readings are historically extreme.

Interbank & Repo Funding Markets

The 3-Month CP-Treasury spread — AA Financial Commercial Paper rate minus the 3-Month Treasury — is the modern successor to the discontinued TED spread. It measures the unsecured short-term credit premium banks pay above the risk-free rate. Readings above 100bps have historically signaled elevated interbank credit stress. SOFR tracks overnight secured repo conditions; divergence from Fed Funds signals repo market pressure.

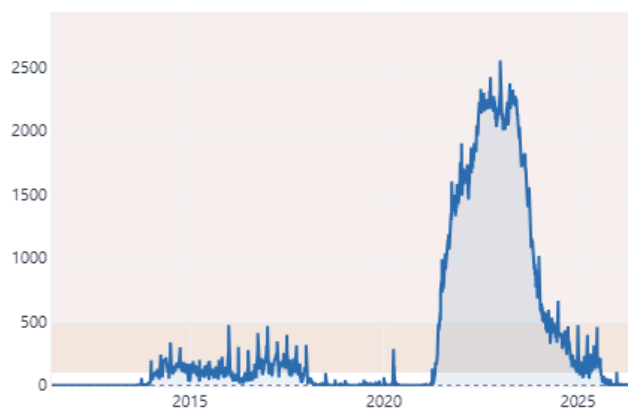
3-Month CP minus Treasury Spread % (Modern TED Equivalent)



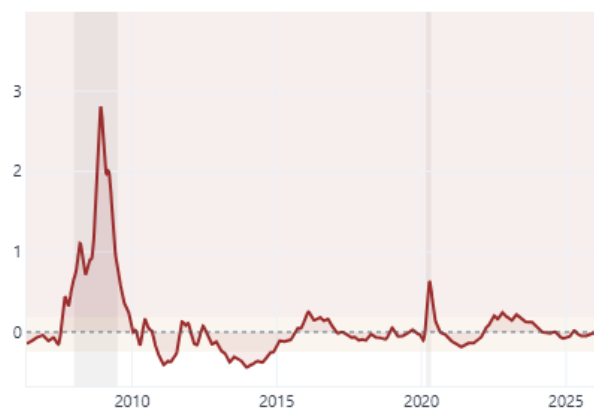
Secured Overnight Financing Rate (SOFR) %



Fed Overnight Reverse Repo Facility (\$B)



NFCI Credit Subindex (Chicago Fed) (current: 54th percentile)



Reverse repo usage peaked near \$2.5 trillion in 2023 as excess reserves flooded the system. Rapid drawdown toward zero — if accompanied by rising funding rates — would signal reserves leaving the system faster than the Fed can replace them. The NFCI credit subindex isolates the credit channel from broader financial conditions, providing an early warning of tightening access to credit specifically.

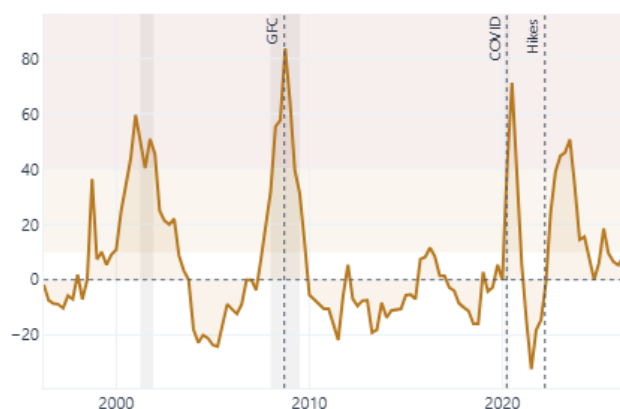
CREDIT MARKET FUNCTIONALITY

Most systemic crises emerge when refinancing capacity deteriorates and credit spreads widen persistently, impairing liquidity and rollover financing. Credit market functionality is a more reliable leading indicator of economic stress than equity market volatility — spreads reflect actual lending conditions, not just sentiment.

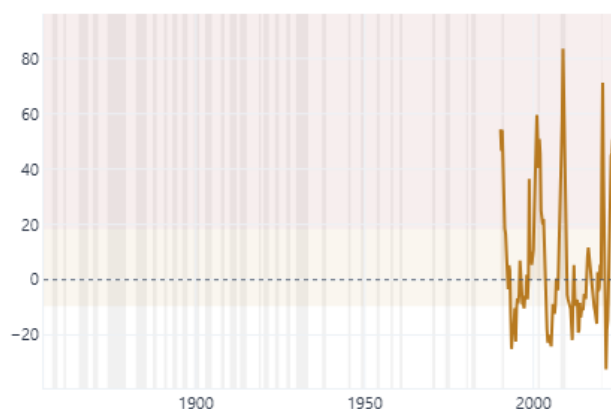
Bank Lending Standards (SLOOS)

The Senior Loan Officer Opinion Survey asks banks whether they tightened or loosened credit standards on C&I; (commercial and industrial) loans in the prior quarter. Net tightening above 40% has historically preceded credit contractions by 6-12 months — making it one of the most reliable leading indicators of future loan growth collapse. It captures supply-side credit rationing before it shows up in delinquencies.

Net % Banks Tightening C&I Loan Standards %



Lending Standards — Historical Percentile (current: 65th percentile)

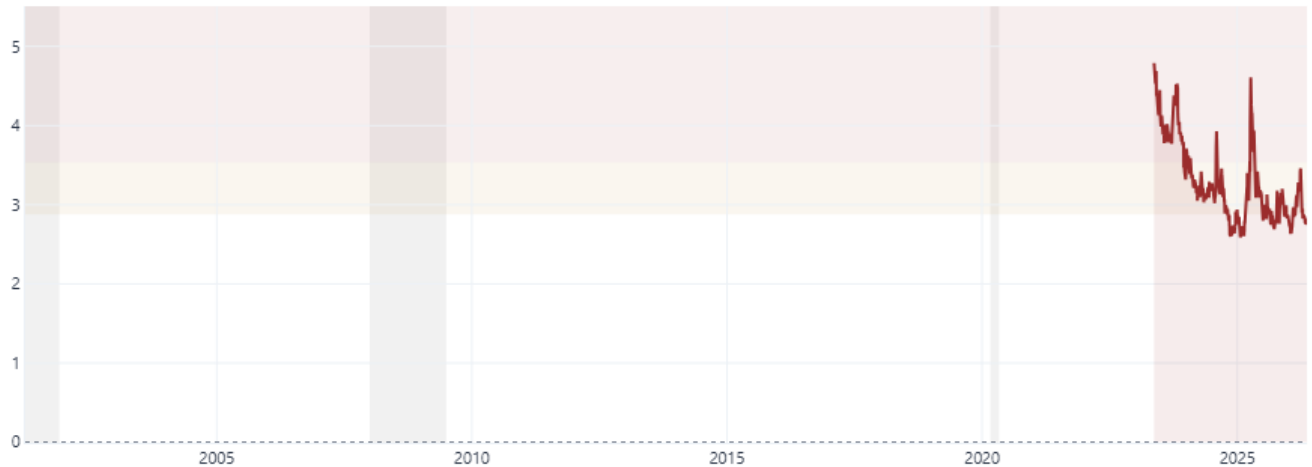


Negative readings (net loosening) indicate accommodative lending conditions. The 2008 GFC and COVID episodes saw rapid swings from loosening to extreme tightening. The current trajectory matters as much as the level — rapidly tightening standards signal banks see rising credit risk ahead of the official data.

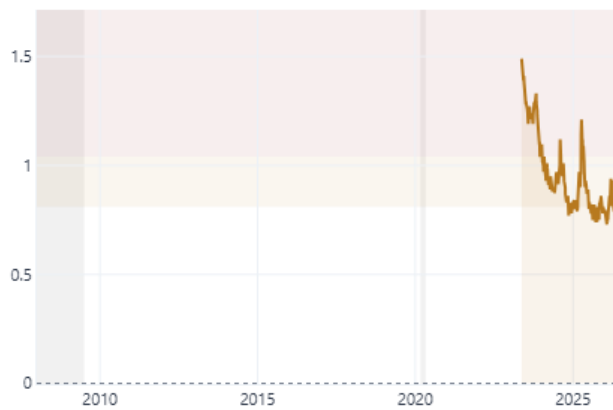
High Yield & Investment Grade Spreads

Credit spreads represent the additional yield investors demand over Treasuries to hold corporate bonds. High yield (HY) spreads are more sensitive to default risk and liquidity; investment grade (IG) spreads reflect credit conditions for larger, more creditworthy issuers. When both widen simultaneously, credit tightening is broad-based.

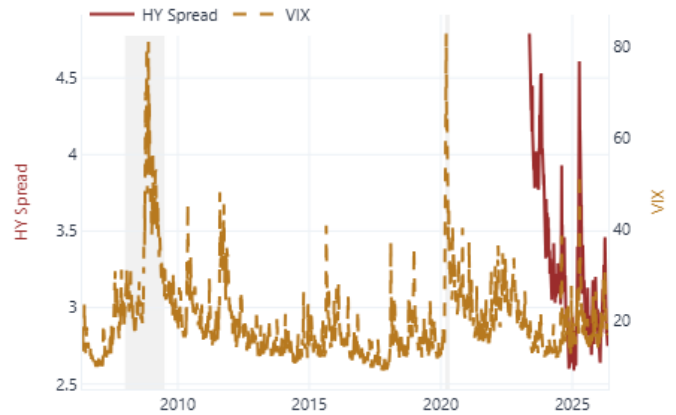
High Yield OAS (ICE BofA) % — Recession-Period Context (current: 10th percentile)



Investment Grade OAS (ICE BofA) % (current: 4th percentile)



HY Spread % (left) vs VIX (right)

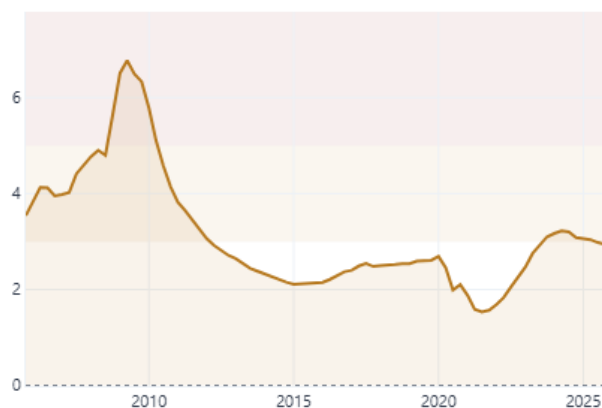


HY spreads and VIX are often correlated in stress episodes but diverge in regime transitions. Persistent spread widening without a corresponding VIX spike suggests structural credit deterioration rather than temporary sentiment-driven volatility — the more concerning scenario for refinancing capacity.

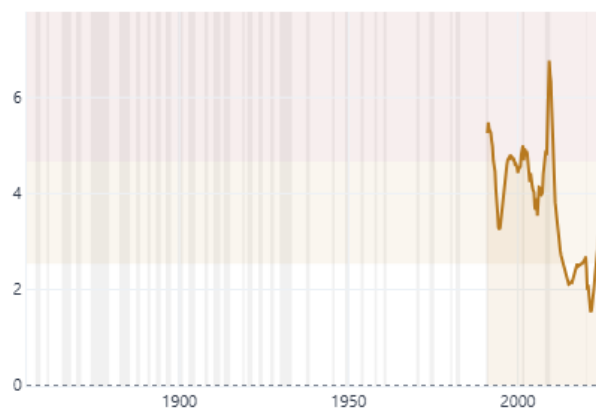
CRE Lending Standards & Consumer Delinquencies

CRE lending standards (SLOOS) capture forward-looking bank risk appetite for commercial real estate — a leading indicator that tightens before delinquencies rise. Credit card delinquencies provide a near-real-time read on consumer balance sheet stress; both are reported quarterly with approximately 6 weeks of lag.

Credit Card Loan Delinquency Rate % (20 Years)

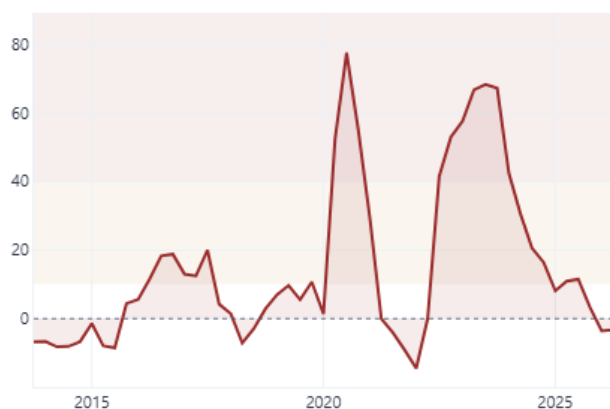


CC Delinquency — Historical Percentile Context (current: 32th percentile)

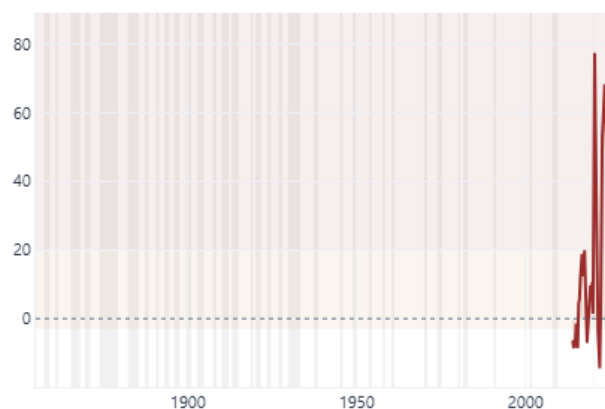


Credit card delinquency is a broad consumer credit health signal. Post-pandemic normalization has pushed this rate above pre-2020 levels — a sign that the lowest-income households are increasingly stretched, even before any broader labor market deterioration.

CRE Lending Standards — Net % Tightening (25 Years)



CRE Lending Standards — Historical Percentile Context (current: 24th percentile)

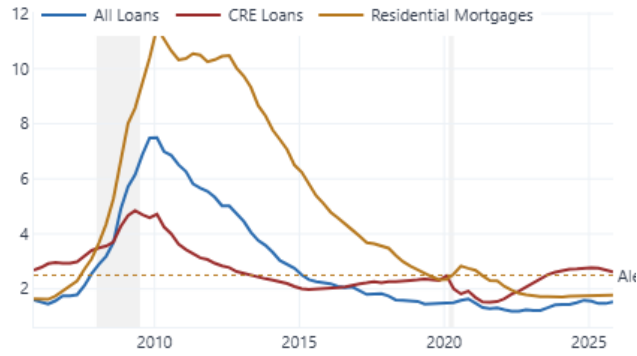


The SLOOS CRE lending standards series captures bankers' forward-looking assessment of commercial real estate credit risk — a leading indicator that typically tightens 6-12 months before delinquency rates rise. Net tightening above 40% has preceded CRE credit contractions in every major stress episode since the 1990 S&L crisis.

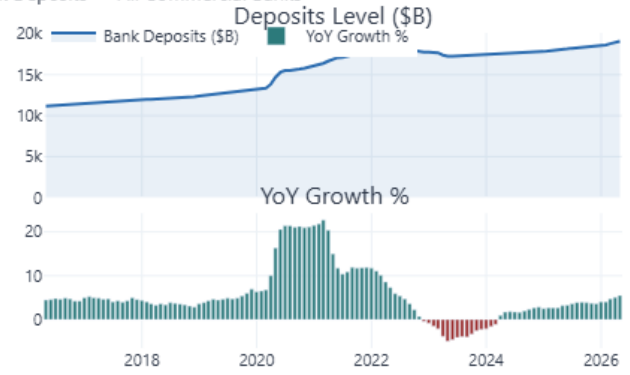
Banking & Credit Quality

Loan delinquency rates across all segments provide a lagging but highly reliable signal of credit cycle stress. All-loan delinquency (blue) is the broadest gauge; CRE (red) is concentrated at regional/community banks; residential mortgages (amber) reflect consumer housing balance sheets. Deposit flows show system-wide funding stability.

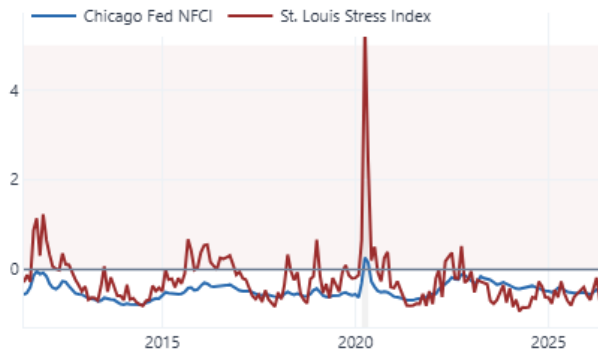
Bank Loan Delinquency Rates — All Loans, CRE, Mortgage



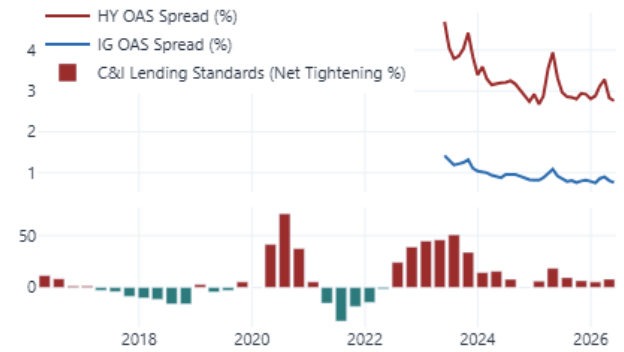
Bank Deposits — All Commercial Banks



Financial Conditions — NFCI & St. Louis FSI



Credit Spreads & Lending Standards — Financial Tightening Channel



The composite financial conditions comparison (NFCI + STLFSI4) shows when overall financial system stress is tightening or loosening. Credit spread overlay with lending standards shows the dual tightening channel — market pricing and bank underwriting simultaneously contracting.

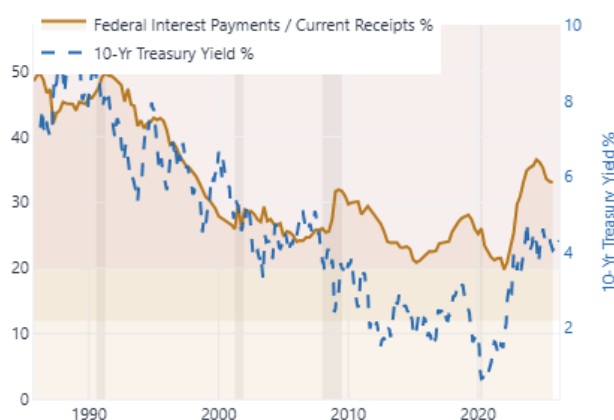
POLICY CONSTRAINTS & FLEXIBILITY

The critical issue is not absolute debt levels, but whether policymakers retain the flexibility to stabilize markets without destabilizing inflation, funding markets, or sovereign confidence. This section tracks the three primary constraints on policy maneuverability: fiscal space (debt service burden), monetary space (inflation expectations), and balance sheet capacity (Fed asset size).

Fiscal Maneuverability: Debt Service Burden

Federal interest payments as a share of current receipts measures how much of government revenue is consumed by debt service. Unlike debt/GDP (which depends on GDP growth assumptions), this ratio directly reflects the cash flow constraint on fiscal policy. Historically below 15%; the post-2022 rate normalization has driven this ratio materially higher as low-rate debt rolls over at current market rates.

Federal Interest Payments / Current Receipts %



Fed Balance Sheet as % of Nominal GDP



Left: interest/receipts ratio (amber) with 10-year Treasury yield overlaid (dashed blue, right axis). The yield overlay shows how the post-2022 rate normalization directly drove the step-change in debt service cost as low-rate debt rolled over at current market rates. Green band: below 12%; yellow: 12-20%; red: above 20%. Right: Fed balance sheet as % of GDP — scale of prior QE and trajectory of QT.

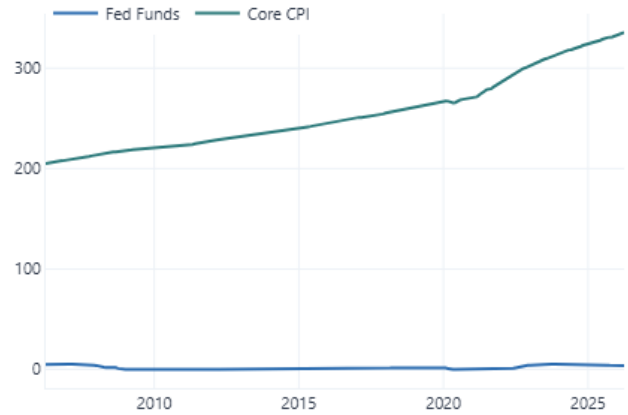
Monetary Constraints: Real Policy Rate & Inflation Expectations

The real Federal Funds Rate (nominal rate minus core CPI YoY) determines whether monetary policy is stimulative or restrictive in inflation-adjusted terms. Deeply negative real rates incentivize borrowing and asset speculation; very high positive real rates risk demand destruction and debt deflation. Inflation expectations are the forward-looking constraint: if markets expect inflation to remain elevated, the Fed cannot reduce rates without undermining its credibility.

Real Federal Funds Rate (FEDFUNDS – Core CPI YoY) %



Federal Funds Rate vs Core CPI YoY % (20 Years)



5-Year Breakeven Inflation Expectations %



10-Year Breakeven Inflation Expectations %



Breakeven rates derived from TIPS. The 2% Fed target implies a sustainable range of approximately 2-2.5% for long-run breakevens. Persistent readings above 3% would indicate inflation expectations becoming unanchored — at which point the Fed would be unable to cut rates in response to economic weakness without risking a credibility crisis. This is the binding constraint on monetary policy flexibility.

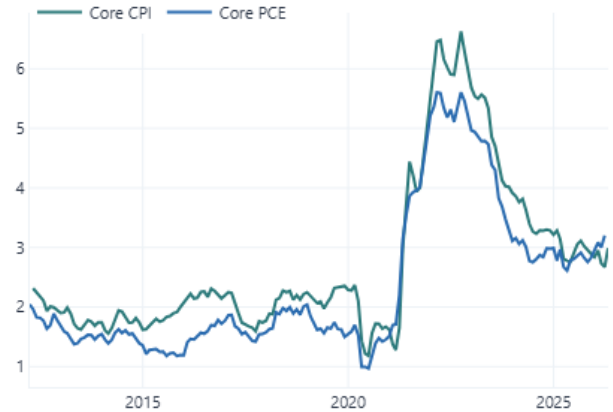
INFLATION

Regime: Elevated | Headline CPI: 3.95% | Core CPI: 2.99% | Core PCE: 3.20% | Median CPI: 4.93% | Trimmed Mean PCE: 2.36% | Supercore: 3.03% | Sticky CPI: 3.04% | OER: 3.65% | Mich Exp 1Y: 3.80% | 5Y Breakeven: 2.70% | 10Y Breakeven: 2.49%

Headline CPI — Year-over-Year %



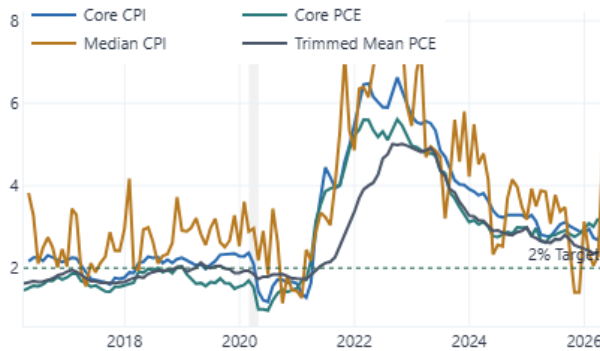
Core CPI vs. Core PCE — YoY %



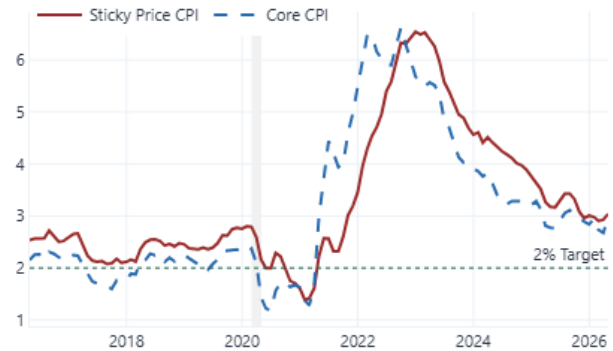
Alternative Inflation Measures

Median CPI (Cleveland Fed) and Trimmed Mean PCE (Dallas Fed) filter extreme price movements and are better predictors of future inflation than Core CPI. Sticky Price CPI captures components that change infrequently — the best predictor of persistent inflation and underlying regime shifts.

Inflation Gauges — Core CPI / PCE, Median CPI, Trimmed Mean PCE



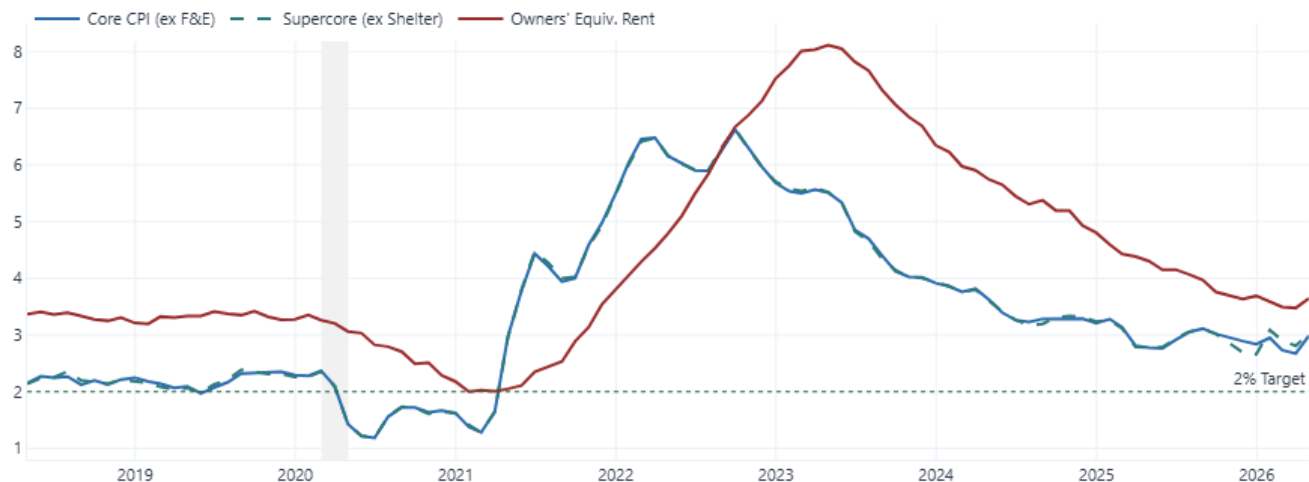
Sticky Price CPI vs. Core CPI — Persistent vs. Transitory



Shelter Decomposition

Owners' Equivalent Rent (OER, ~26% of CPI) lags actual market rents by 12–18 months. Supercore CPI (Core ex-Shelter) shows services inflation driven by wages — the Fed's primary focus for determining when disinflation is structural vs. transitory.

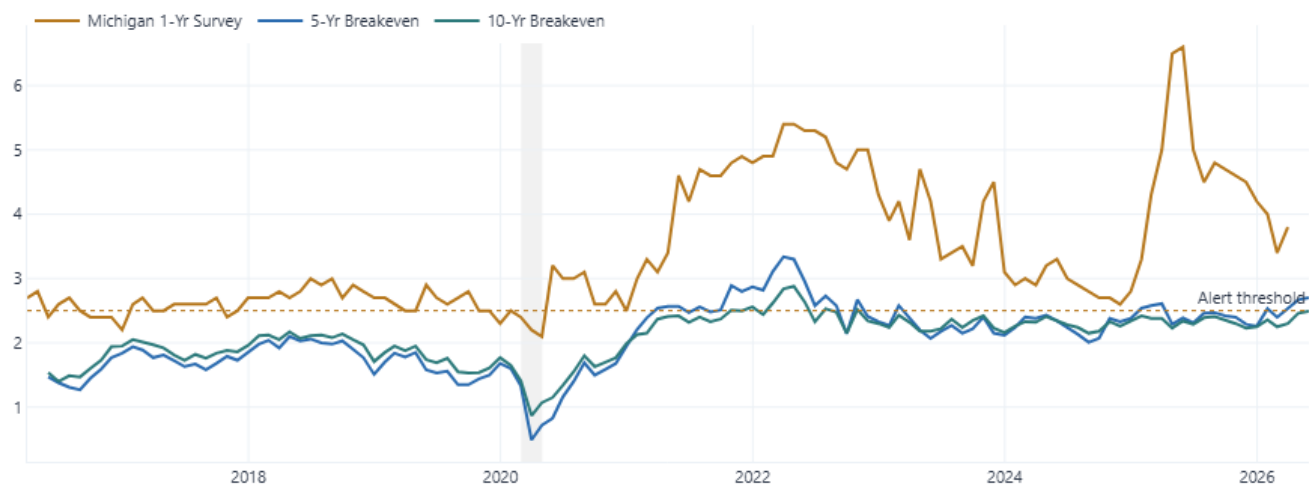
Inflation Decomposition: Core CPI vs. Supercore vs. OER



Inflation Expectations

Well-anchored expectations are essential to the Fed's credibility. Michigan 1-year survey persistently above 4% or 5/10-year breakevens above 2.5% signal deanchoring risk that historically requires more aggressive policy response.

Inflation Expectations — Survey & Market Breakevens



Money Supply

M2 Growth — Year-over-Year %



Real M2 Growth — Year-over-Year %



Rapid M2 growth historically precedes inflation with a 12–18 month lag. Real M2 contraction signals tightening monetary conditions that typically suppress demand-driven price pressure within 2–4 quarters.

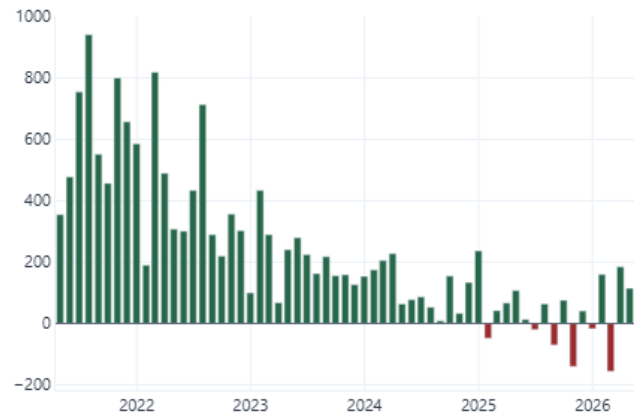
LABOR MARKET

Labor Deterioration Index: 44/100 | U-3: 4.3% | U-6: 8.2% | U6-U3 Gap: 3.9pp | AHE YoY: 3.6% | Real Wage: -0.4% | ULC YoY: 1.2% | Quits: 2.00% | Layoffs: 1.20%

Unemployment: U-3 vs U-6 (Top) | Slack Gap (Bottom)



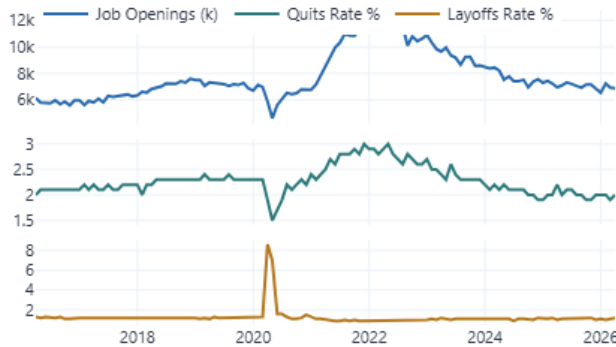
Nonfarm Payrolls — Monthly Change (Thousands)



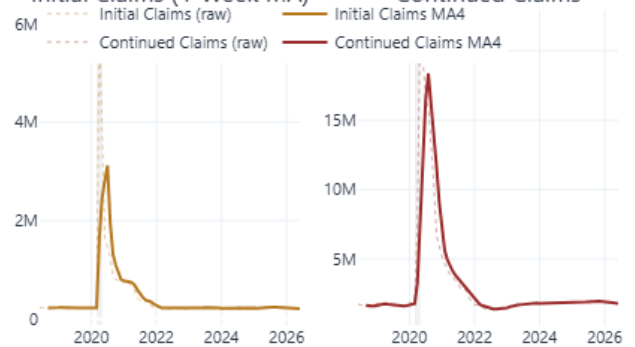
JOLTS & Jobless Claims

Job openings lead payroll changes by 2–4 months. The quits rate falls when workers lose confidence in job mobility — a reliable 3–6 month leading indicator of rising unemployment. Continued claims measure re-hiring absorption.

JOLTS: Openings / Quits / Layoffs



Jobless Claims — Initial & Continued



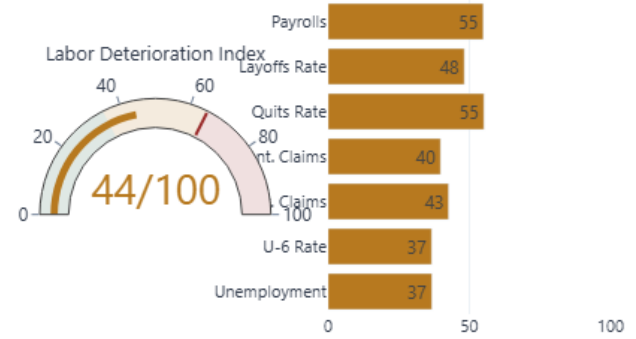
Wages, Productivity & Unit Labor Costs

Unit Labor Costs (wages ÷ productivity) is the primary transmission mechanism from labor costs to goods and services prices. Sustained ULC above 2.5% YoY is inconsistent with the Fed's 2% inflation target. ECI YoY: 3.4% | Productivity YoY: 2.9%.

Wages vs. Productivity vs. Unit Labor Costs — YoY %

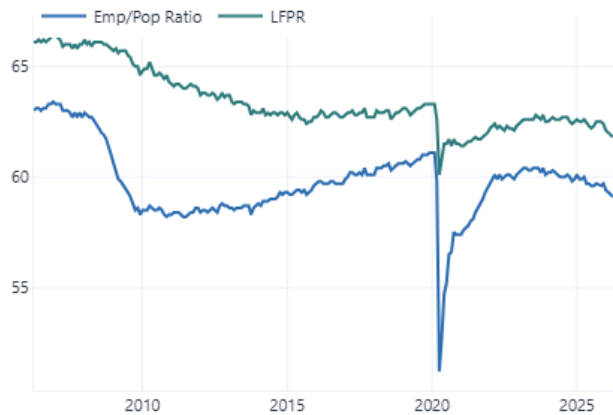


Labor Deterioration Index — Current Reading & Components



Employment Structure & Industrial Activity

Employment-Population Ratio vs. Participation Rate



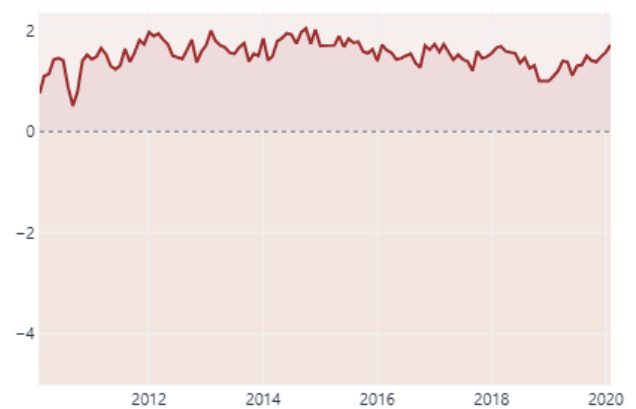
Industrial Production — Year-over-Year %



Personal Saving Rate %



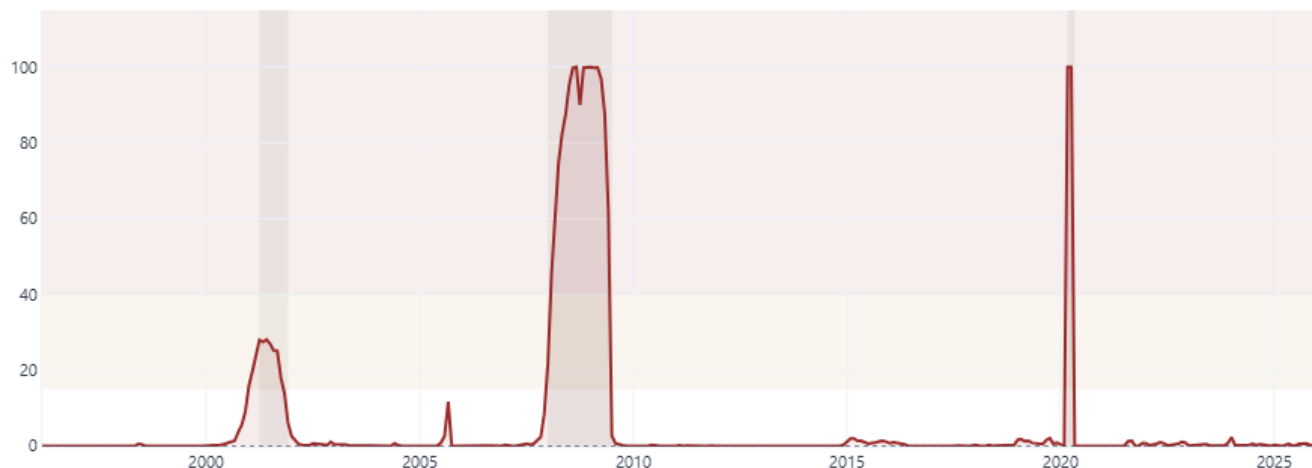
Conference Board Leading Economic Index (MoM %)



Industrial production YoY contraction has coincided with every U.S. recession since 1960. The Conference Board LEI aggregates 10 forward-looking components; three consecutive monthly declines have preceded every NBER recession with a 6–12 month lead.

MARKETS & RATES

NY Fed 12-Month Recession Probability %



The NY Fed recession probability is derived from the yield curve (3-month vs 10-year spread). Readings above 30-40% have historically signaled recession within 12 months.

10-Year Treasury Yield — Daily vs. Monthly Average %



Broad Dollar Index — YoY % Change



A sharply rising dollar (>8% YoY) acts as a global tightening mechanism: it raises debt service costs for dollar-denominated EM borrowers, pressures commodity prices, and reduces earnings for US multinationals. Dollar strength in a late-cycle environment amplifies the restrictive effect of Fed policy beyond US borders.

S&P 500 CAPE (Shiller P/E) — Full History

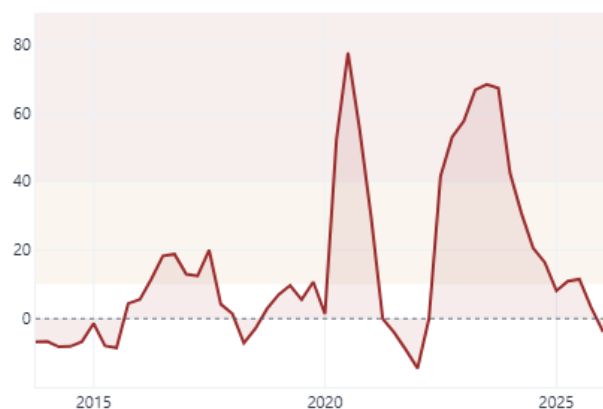


The Cyclically Adjusted P/E ratio smooths earnings over a rolling 10-year real average, removing business cycle distortions. The historical mean is approximately 17. Readings above 30 — sustained through 1929, 2000, and 2021 — have historically preceded decade-long below-average real returns. CAPE does not predict timing.

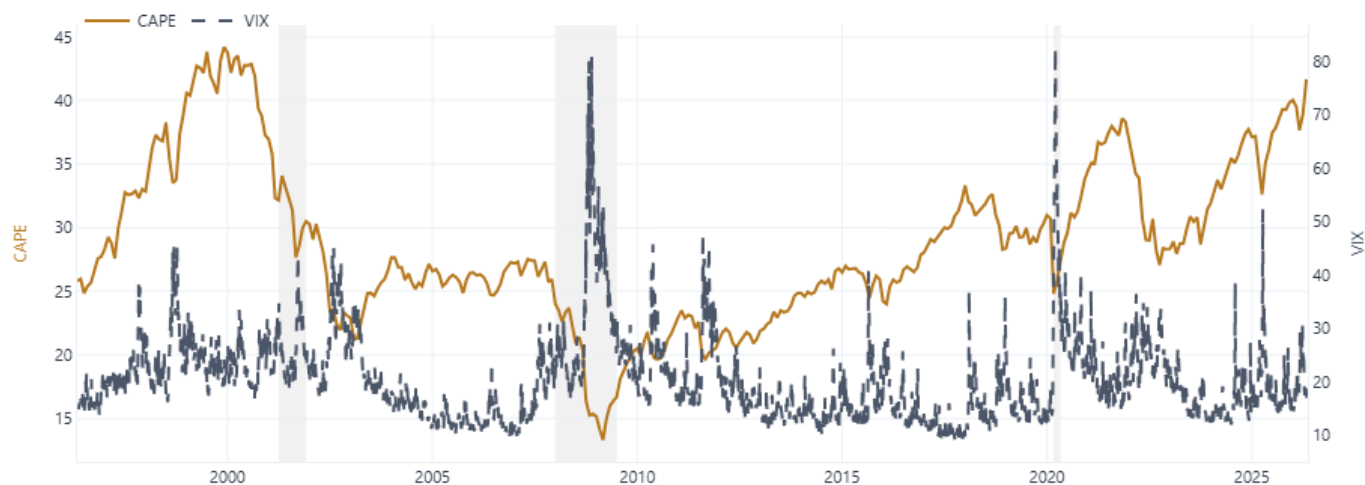
S&P 500 Trailing P/E (40 Years)



CRE Lending Standards — Net % Tightening



S&P 500 CAPE (Shiller P/E, left) vs VIX Volatility Index (right)



CAPE measures structural valuation risk (where valuations are relative to long-run earnings); VIX measures near-term sentiment and implied volatility. High CAPE is a long-run forward return predictor, not a short-term timing signal — elevated readings have coexisted with extended bull markets. Low VIX reflects current market calm and can persist. The combination most associated with historical drawdowns is high CAPE combined with a VIX spike above 30 (2001, 2008, 2022) — both elevated valuations and materialising stress, not valuations alone.

Yield Curve: 10-Year minus 2-Year Treasury Spread %



The yield curve spread (10Y minus 2Y) is arguably the most reliable leading recession indicator: inversion has preceded every US recession since 1970, typically by 12-18 months. Red zones mark periods of inversion (spread < 0). Note that uninverting after a prolonged inversion can coincide with — rather than precede — economic contraction.

10-Year Treasury Yield — Long History %



10-Year Treasury Yield — 10 Years



The secular decline in long-term yields from ~16% in 1981 to near 0% in 2020 supported rising asset valuations and reduced debt service costs. The reversal since 2022 has structural implications for real estate valuations, corporate refinancing, and government debt service.

HOUSING MARKET

Housing is the most interest-rate-sensitive sector in the economy and a critical transmission mechanism for monetary policy. Mortgage rates directly control affordability; starts lead construction employment by 3-6 months; and home prices affect household net worth, consumer confidence, and collateral values for lending.

Mortgage Rates & Housing Starts

30-Year Fixed Mortgage Rate %



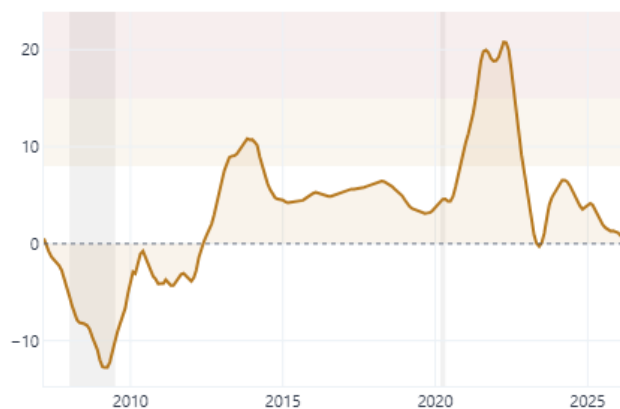
Housing Starts — Total (Thousands, SAAR)



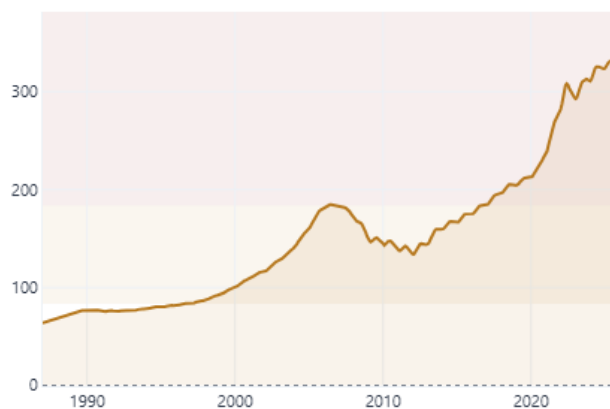
The 30-year mortgage rate rose from sub-3% in 2021 to over 7% in 2023 — the fastest rate increase in the modern mortgage market era. Housing starts respond with a lag of 3-6 months to rate changes, as pipeline projects run off. The secular housing undersupply (estimated 3-5M units nationally) provides a structural floor that limits the depth of a housing correction compared to 2008.

Home Price Appreciation

Case-Shiller Home Price Index — YoY %



Case-Shiller HPI — Historical YoY Percentile (current: 98th percentile)



Home price appreciation above 8% YoY outpaces wage growth and compresses affordability faster than income can adjust. The 2020-2022 surge — driven by pandemic-era demand shifts and 3% mortgages — has proved partially sticky even as rate headwinds increased. Elevated home prices combined with high mortgage rates create an 'affordability lock': would-be sellers are reluctant to trade their 3% mortgage for 7%, suppressing transaction volume.

Mortgage Rate % vs. Housing Starts (Thousands) — 20 Years



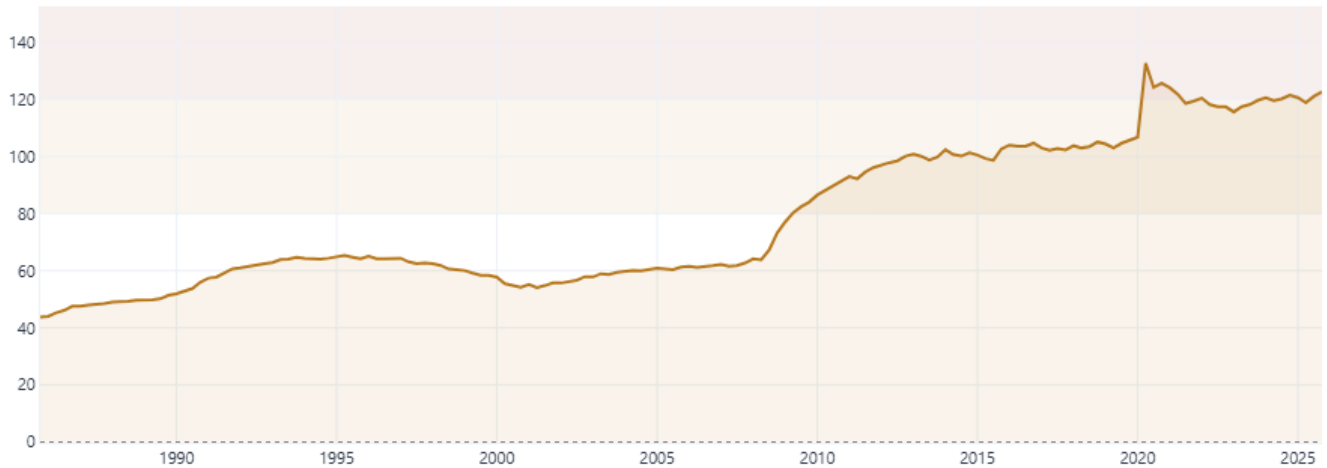
30-Year Mortgage Rate — 40-Year History %



The 40-year history chart provides crucial context: the 1980s featured mortgage rates above 15%. The post-2008 secular decline to sub-3% was a structural tailwind for home prices and homeowner net worth. The reversal since 2022 is large in absolute terms but still well below historical peaks.

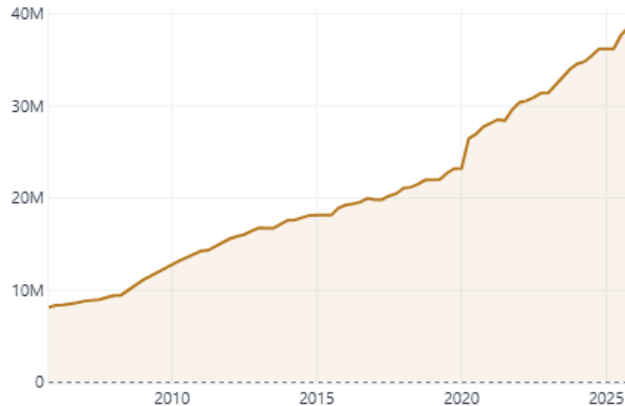
FISCAL

Federal Debt as % of GDP

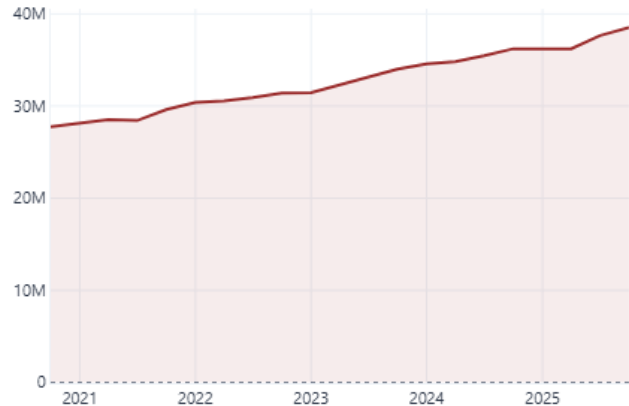


Federal debt as a share of GDP rose sharply during World War II (peaking ~106%), declined through the postwar expansion, then began climbing again from the 1980s. The 2008 financial crisis and the 2020 pandemic triggered the largest peacetime increases on record. At elevated debt/GDP ratios, interest rate increases have a compounding effect: as debt rolls over at higher rates, the share of the budget consumed by interest payments expands, crowding out discretionary fiscal space.

Federal Debt Outstanding (\$B, Monthly)

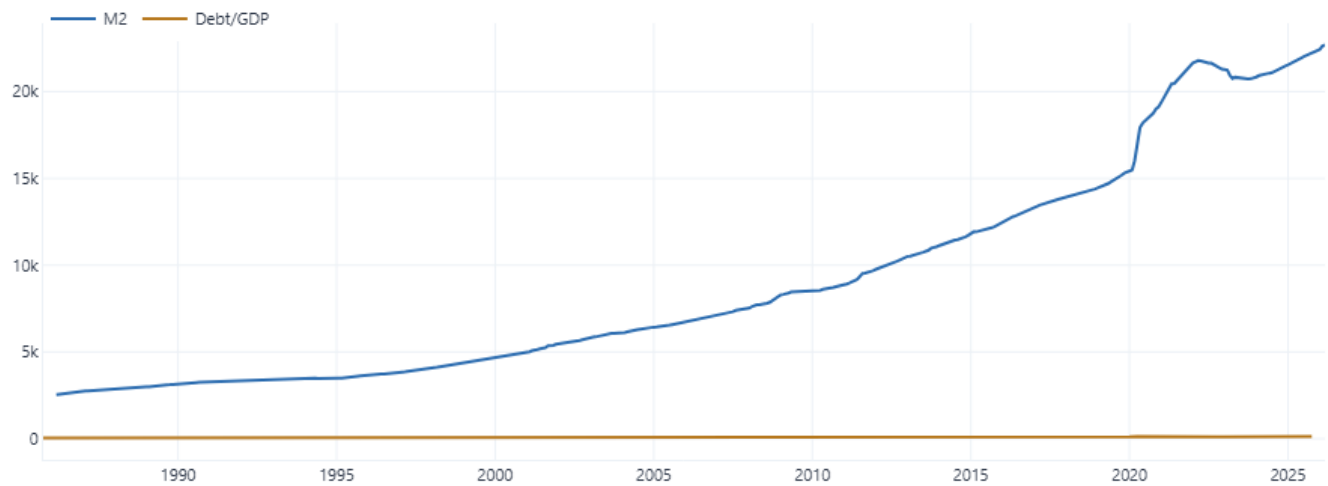


Federal Debt Outstanding (\$B) — Last 5 Years



Monthly total public debt outstanding provides a more timely read than the quarterly Debt/GDP ratio, which lags by a full quarter. The 5-year view highlights the pace of recent accumulation relative to the long-run trend.

M2 Money Supply (B\$) & Federal Debt/GDP — Long-Run Trends



Note: M2 is shown in billions of dollars; Debt/GDP is a percentage. The two series share a chart to illustrate the co-movement of monetary and fiscal expansion over the past four decades.

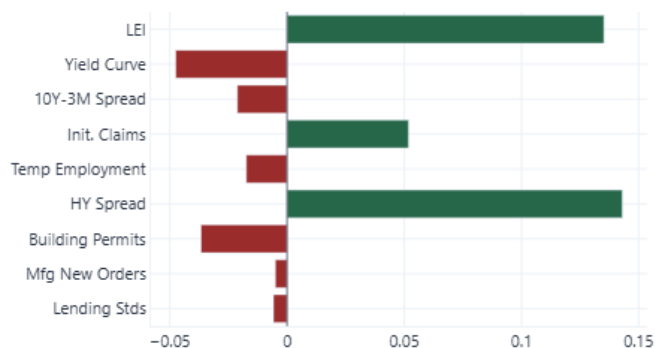
LEADING INDICATORS & BUSINESS CYCLE INDEX

The Composite Business Cycle Index (BCI) aggregates eight leading macro indicators into a single standardized score. Each component is z-scored against its 20-year calibration window, inverted when lower values are healthier, then combined as a weighted average (weights renormalized for missing data) and smoothed with a 3-month moving average. **Current BCI: 0.09** — **Phase: Expansion**

Composite Business Cycle Index (BCI)



BCI Component Contributions (Current)



Expansion	BCI > 0, momentum positive — above-trend growth accelerating
Slowdown	BCI > 0, momentum negative — above-trend but decelerating
Recovery	BCI ≤ 0, momentum positive — below-trend but improving
Contraction	BCI ≤ 0, momentum negative — below-trend and worsening

Trend & Momentum Analytics

Direction is based on the 3-month linear trend: **Deteriorating** means the series is trending toward worse conditions; **Improving** means improving. Z vs 1Y measures how far the current reading departs from its 12-month average in standard-deviation units.

Series	Current	3M Trend	6M Trend	Accel.	Z vs 1Y	Direction
LEI	1.72	+0.120	+0.048	+0.072	+1.99σ	Improving
Yield Curve	0.50	-0.005	-0.051	+0.046	-0.72σ	Stable
10Y-3M Spread	0.90	+0.150	+0.075	+0.075	+1.72σ	Improving
Init. Claims	211000.00	+4000.000	-2600.000	+6600.000	-0.42σ	Deteriorating
Temp Employment	-1.68	+0.367	+0.374	-0.007	+1.75σ	Improving
HY Spread	2.76	-0.260	-0.007	-0.253	-1.05σ	Improving
Building Permits	-7.97	-1.450	+0.113	-1.562	-1.05σ	Deteriorating
Mfg New Orders	2.11	-1.069	-0.155	-0.914	-0.79σ	Deteriorating
Lending Stds	8.10	+0.800	-0.946	+1.746	-0.46σ	Deteriorating

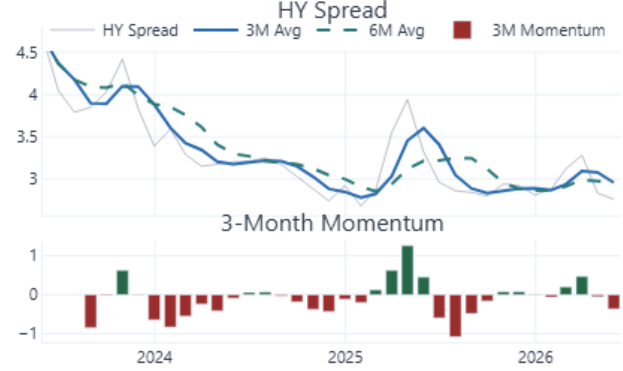
Indicator Trend Details

Top panel: indicator level (faint) with 3-month (solid blue) and 6-month (dashed teal) rolling averages. Bottom panel: 3-month rate of change — green bars indicate improving momentum, red bars indicate deterioration.

Leading Economic Index — Trend & Momentum



High Yield OAS (ICE BofA) — Trend & Momentum



Yield Curve Spread (10Y - 2Y) — Trend & Momentum



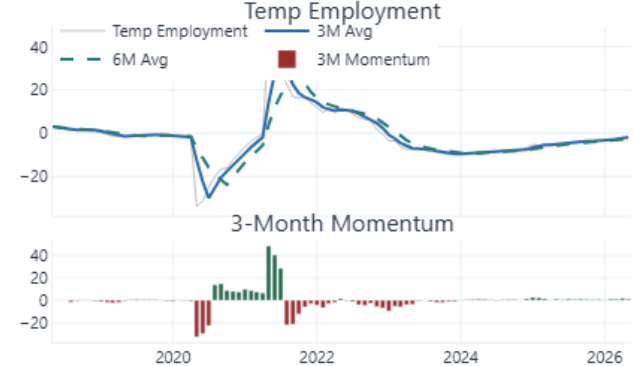
Initial Jobless Claims — Trend & Momentum



Building Permits (New Private Housing) — Trend & Momentum



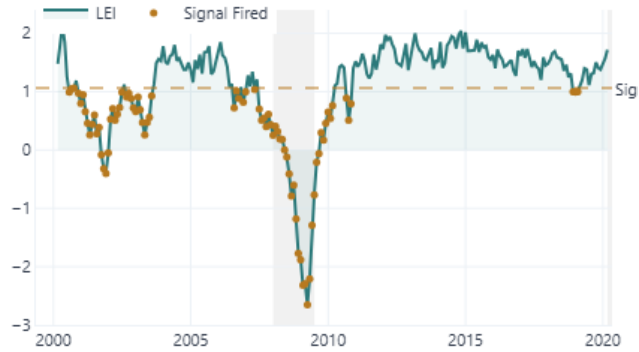
Temporary Help Services Employment — Trend & Momentum



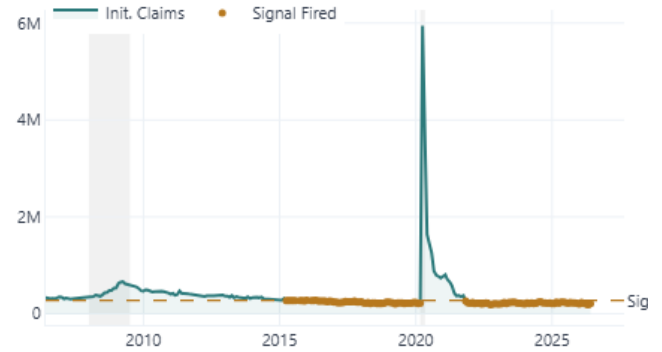
Historical Signal Overlay Charts

Each chart shows the indicator series with NBER recession shading (grey), the signal threshold (dashed amber line at the 75th or 25th percentile of the full history), and orange dots marking where the signal fired. Visual validation of hit rates and false-positive rates.

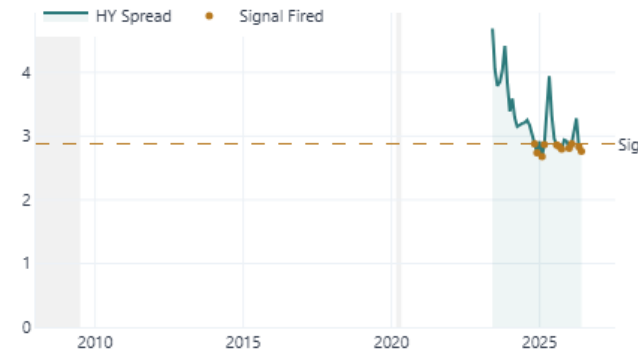
LEI — Historical Signal vs. NBER Recessions



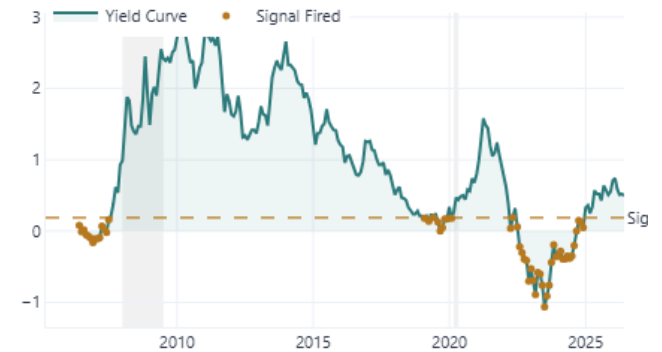
Init. Claims — Historical Signal vs. NBER Recessions



HY Spread — Historical Signal vs. NBER Recessions



Yield Curve — Historical Signal vs. NBER Recessions



Historical Recession Validation

Hit rate: percentage of NBER recessions (within the indicator's data history) preceded by a stressed signal — series above the 75th percentile (`higher_is_bad=True`) or below the 25th percentile (`higher_is_bad=False`) — within 18 months before the recession start. Lead time: average months of first advance warning (first signal in the window). False positive rate: percentage of signal episodes not followed by a recession within 24 months. Only recessions within each indicator's data history are evaluated; indicators with insufficient history are omitted.

Series	Hit Rate	Avg Lead (Mo.)	False Pos. Rate	Recessions
LEI	100%	14.3	33%	4
Yield Curve	100%	17.2	40%	6
10Y-3M Spread	100%	16.0	60%	4
Init. Claims	25%	11.0	95%	8
Temp Employment	67%	3.5	67%	3
Building Permits	88%	12.6	65%	8
Mfg New Orders	67%	9.5	80%	3
Lending Stds	75%	6.4	57%	4

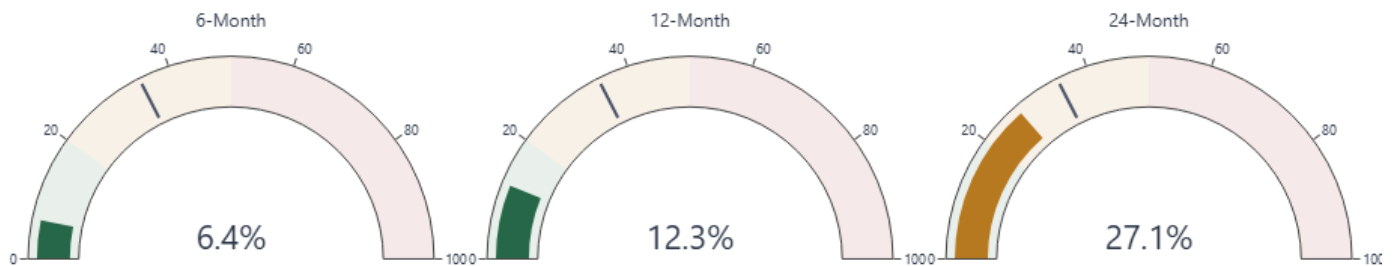
Note: Statistics reflect in-sample backtesting against NBER post-hoc recession dating and are limited to recessions within each indicator's available data history. ICE BofA HY/IG spread series (BAMLH0A0HYM2) are omitted because FRED's API returns only ~2 years of history due to licensing restrictions — insufficient for meaningful backtest evaluation. Initial jobless claims (ICSA) is a coincident indicator and not expected to signal recessions 18 months in advance. Data revisions may introduce mild lookahead bias. These figures should inform, not replace, judgement.

RECESSION PROBABILITY MODEL

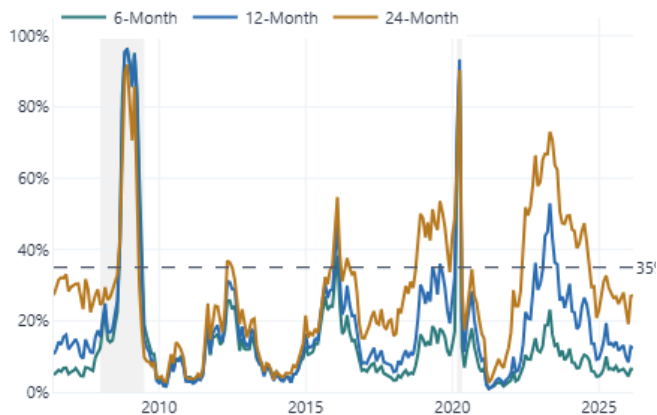
6M: 6.4% (Low) 12M: 12.3% (Low) 24M: 27.1% (Elevated)

Recession probabilities are estimated by a pure-numpy logistic regression model trained on NBER recession dates using eight macro features: yield curve (2Y and 3M spreads), Conference Board LEI (YoY), initial jobless claims (log, 4-week MA), ISM manufacturing new orders, high-yield credit spreads, unemployment rate change, and payroll growth. Separate models are trained per horizon (6M, 12M, 24M) with L2 regularisation. A 35% probability threshold is used for recession signal flags.

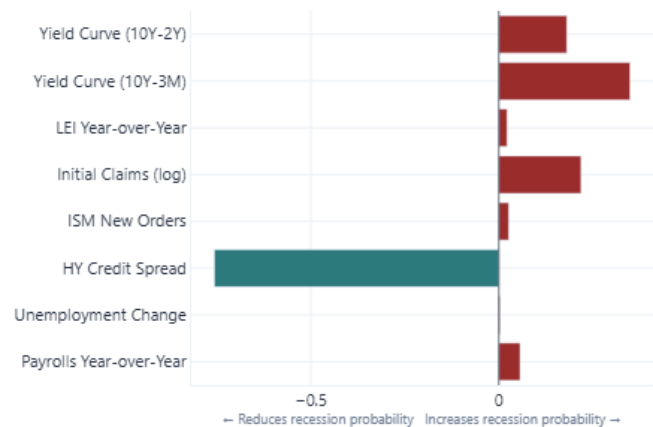
Recession Probability by Horizon



Recession Probability — All Horizons



Feature Contributions — 12-Month Recession Probability



Out-of-Sample Backtest Performance

Rolling out-of-sample backtest: the model is re-trained each month using only data available at that time (20-year window), then tested one step ahead. Precision = fraction of high-probability signals followed by a recession. Recall = fraction of recessions preceded by a high-probability signal. Hit rate = recessions with at least one signal fired within the horizon.

Horizon	Precision	Recall	Hit Rate	False Pos. Rate	Obs.
6M	72%	70%	92%	N/A	0
12M	61%	74%	86%	N/A	0
24M	57%	90%	78%	N/A	0

Note: Logistic regression is a baseline model. It captures linear relationships between macro features and recession risk but does not account for structural breaks, policy regime changes, or non-linear interactions. Probabilities should be interpreted as a signal of elevated risk, not a precise forecast.

RISK TAXONOMY SCORECARD

Risk is organised into four conceptually distinct categories that can decouple from each other during different macro regimes. Mixing them produces misleading aggregate scores — a benign credit environment can mask cyclical deterioration, and fiscal imbalances can persist for years before affecting near-term growth.

Cyclical Recession Risk — Yellow

Aggregates leading macro-cycle signals that have historically peaked 3–18 months before recession onset: the Conference Board LEI, yield curve slope, NY Fed model probability, jobless claims, ISM manufacturing new orders, and building permits.

Indicator	Current Value	As Of	Risk Level
Leading Econ Index	1.7	1 Feb 2020	Green
Yield Curve (10Y-2Y)	0.5%	15 May 2026	Green
NY Fed Rec. Prob.	1.8%	1 Mar 2026	Green
Initial Claims	211,000.0	9 May 2026	Green
ISM New Orders	2.1%	1 Mar 2026	Yellow
Building Permits	-8.0%	1 Mar 2026	Green

Financial Stability Risk — Green

Credit, funding, and systemic stress indicators independent of the macro cycle. Financial stress can emerge without recession risk, and vice versa. Persistent deterioration here impairs the system's ability to self-stabilize when shocks arrive.

Indicator	Current Value	As Of	Risk Level
Financial Stress Index	-0.8	8 May 2026	Green
Nat'l Financial Cond.	-0.5	8 May 2026	Green
High Yield Spread	2.8%	14 May 2026	Green
Investment Grade Spread	0.8%	14 May 2026	Green
C&I; Lending Standards	8.1%	1 Apr 2026	Green
CRE Lending Standards	-3.3%	1 Apr 2026	Green
CP-Treasury Spread	0.03%	13 May 2026	Green

Valuation & Sentiment Risk — Red

Equity market valuations and implied volatility. Elevated valuations reduce the margin of safety (not a recession predictor) while compressed VIX alongside stretched CAPE indicates valuations are stretched relative to near-term risk pricing — associated with elevated long-run drawdown risk, but not a reliable short-term timing signal.

Indicator	Current Value	As Of	Risk Level
CAPE (Shiller P/E)	41.7	1 May 2026	Red
Trailing P/E	31.9	1 May 2026	Yellow
VIX	17.3	14 May 2026	Green

Fiscal & Policy Risk — Red

Fiscal sustainability and monetary policy constraints. High debt service plus unanchored inflation expectations simultaneously constrain both fiscal and monetary stabilization tools — the combination that historically produces the most severe outcomes.

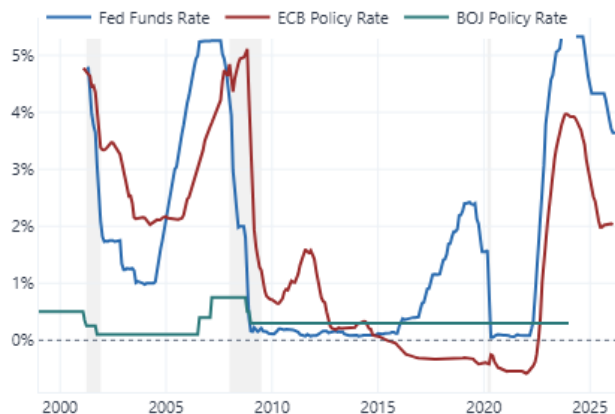
Indicator	Current Value	As Of	Risk Level
Federal Debt / GDP	122.6%	1 Oct 2025	Red
Interest / Gov't Receipts	33.0%	1 Oct 2025	Red
5-Yr Infl. Expectations	2.7%	15 May 2026	Yellow
10-Yr Infl. Expectations	2.5%	15 May 2026	Green
Real Fed Funds Rate	0.7%	1 Apr 2026	Green

Risk levels are assigned by threshold or percentile rules defined per series in series_registry.py. Green = below alert threshold; Yellow = moderate stress; Red = high stress. Overall category score is the maximum component risk level.

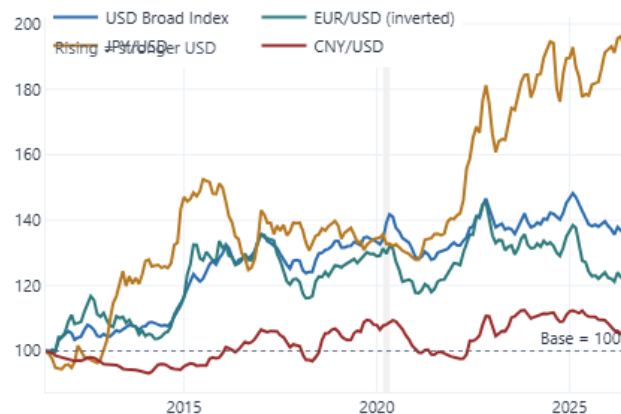
GLOBAL MACRO & FX

Central Bank Divergence: Fed 3.64% | ECB 2.03% | BOJ 0.30% | Fed–ECB gap: +1.61pp — US Rate Advantage · **USD Broad:** -3.3% YoY (Neutral) · **Commodity Pressure:** +38.7% avg YoY — Commodity Surge (Inflationary)

Central Bank Policy Rates — Fed vs ECB vs BOJ



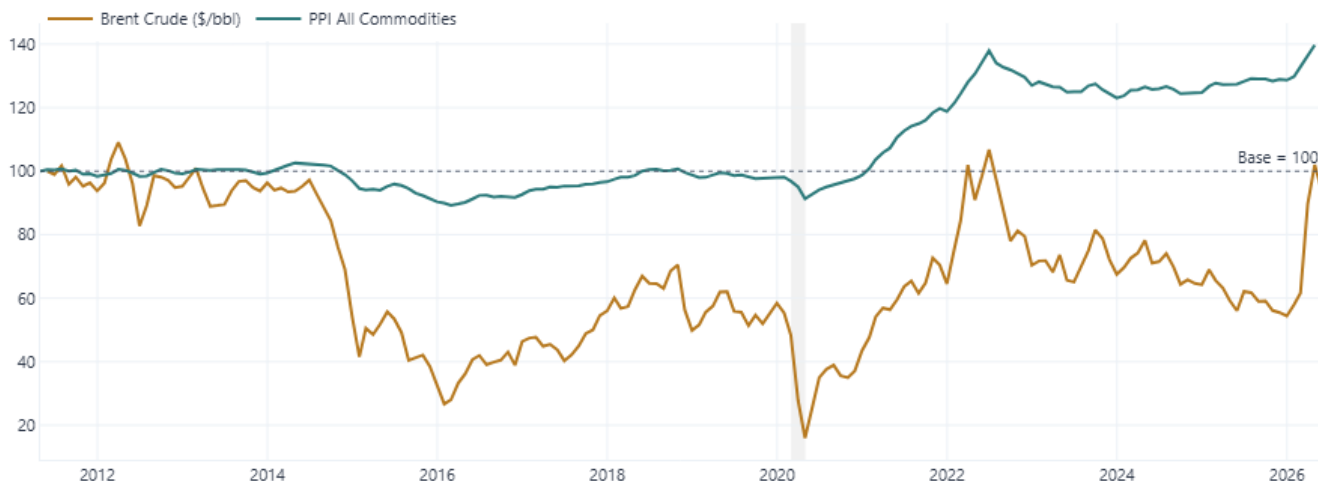
FX Conditions — USD Broad Index & Major Pairs (Indexed to 100)



Global Commodity Complex

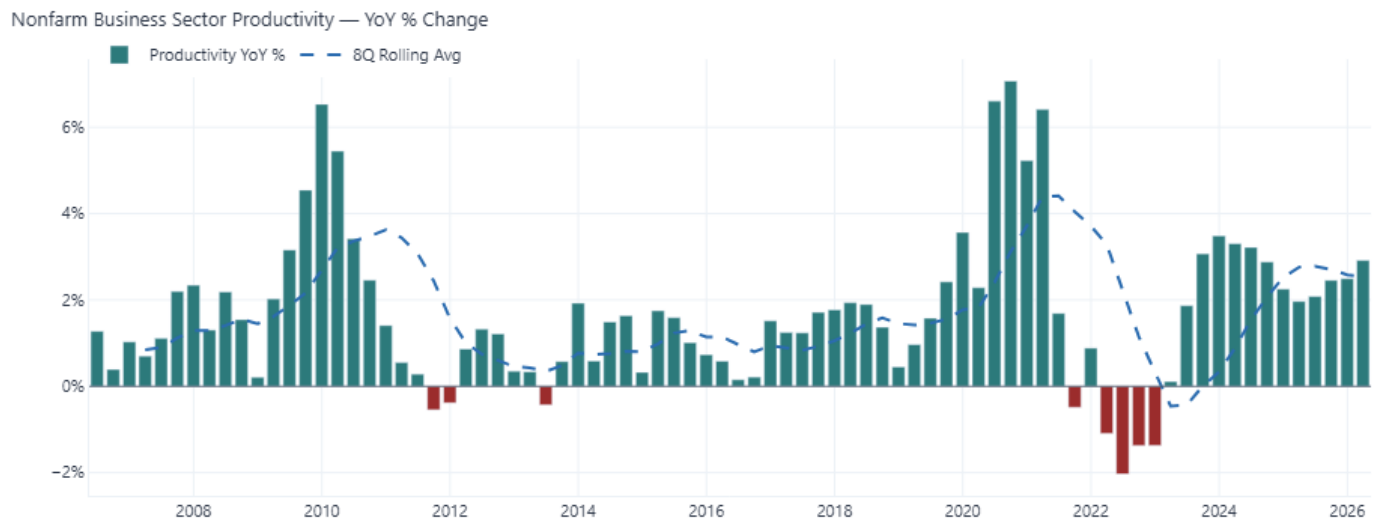
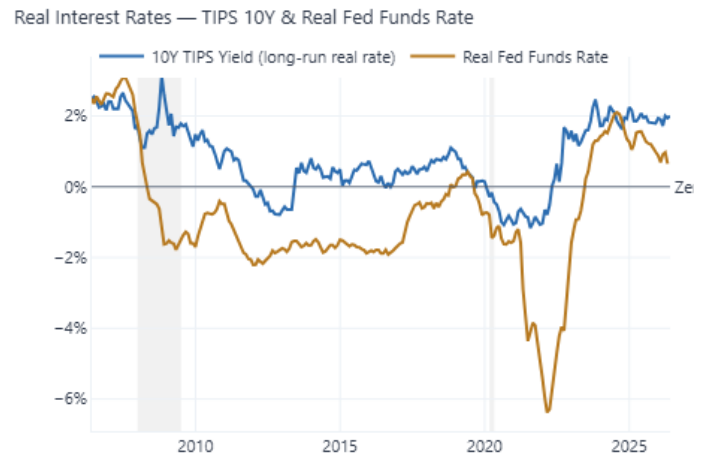
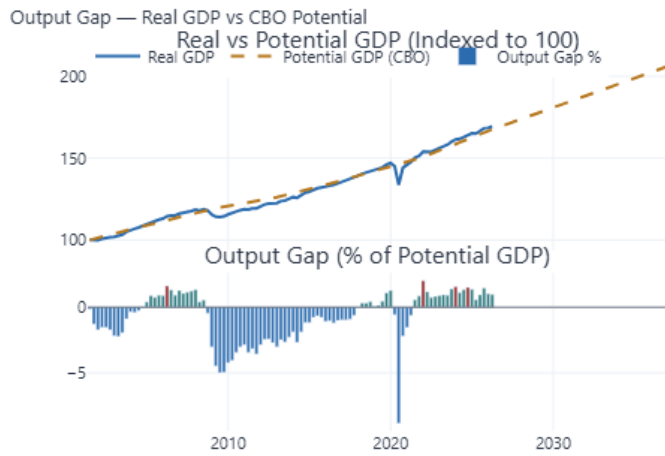
Brent crude: \$108/bbl (+67.5% YoY). Gold: \$N/A/oz (N/A). PPI All Commodities: +9.8% YoY. Commodity surges lead consumer prices by 2–3 months via energy and food pass-through. Gold reflects safe-haven demand and long-run inflation expectations.

Global Commodity Complex — Indexed to 100 at Period Start



Structural & Long-Run Context

Output gap: +0.96% (Above Potential). **Real FF rate:** +0.65% vs 10Y real rate (TIPS proxy) 2.00% (note: TIPS yield includes a real term premium above true r^*) → Stance: **Accommodative**. **Demographics:** Working-age population growth +1.48% YoY — Favorable Demographics. Demographic deceleration structurally constrains potential GDP growth and increases the pension/healthcare fiscal burden over the next decade.



MACRO REGIME CLASSIFICATION

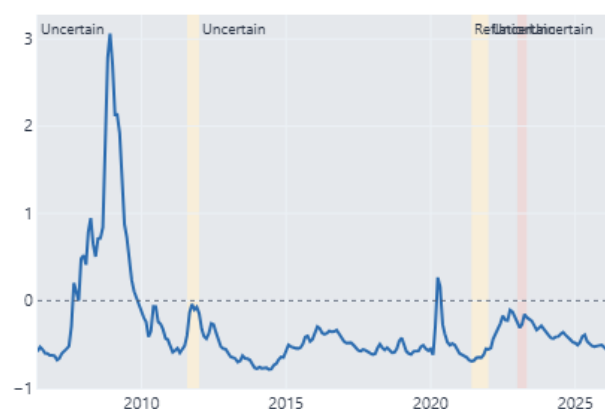
Current Regime: Uncertain · Mixed signals across growth, inflation, and financial conditions dimensions. No single regime has sufficient probability mass to warrant a confident classification.

Dimension	Score	Label
Growth	0.24	Expansion
Inflation	0.55	Elevated
Financial Conditions	0.11	Easy
Credit	0.29	Expanding

Regime Dimension Scores — 0 = Healthy, 1 = Stressed



Macro Regime History



Regime Definitions & Portfolio Implications

Goldilocks: Broad risk-on; equities, credit, cyclicals outperform. Duration neutral.

Reflation: Commodities, energy, financials outperform. Underweight duration.

Stagflation: Real assets, commodities, TIPS. Equities underperform; avoid duration.

Disinflation: Duration outperforms. Quality equities. Avoid commodities.

Liquidity Boom: Risk assets and credit outperform. Watch for late-cycle imbalances.

Tightening Cycle: Underweight duration. Financials mixed. Quality over quantity in credit.

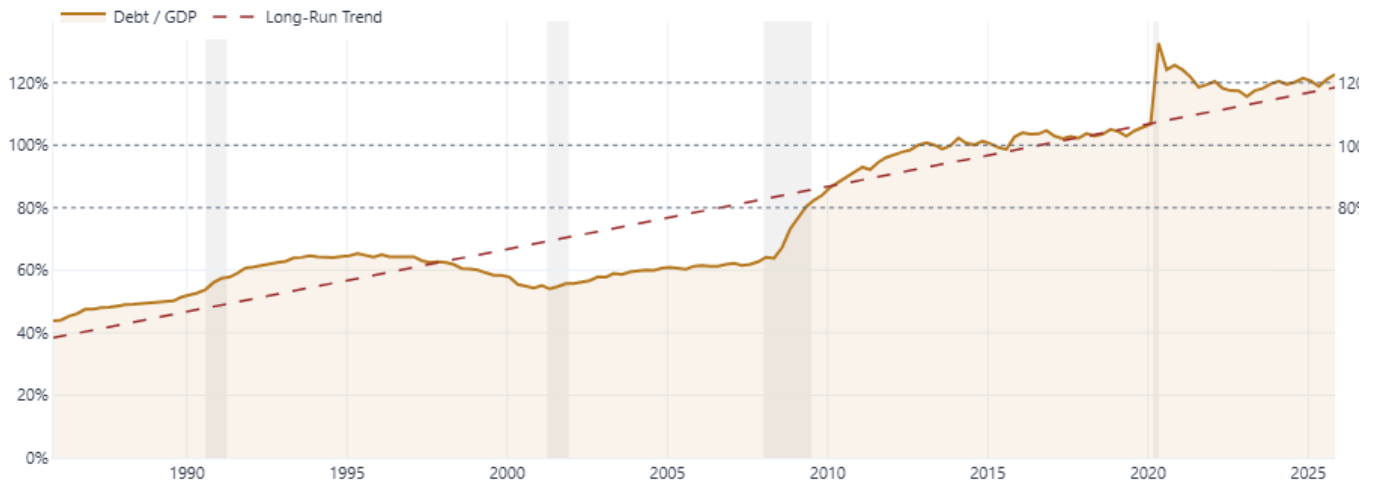
Balance-Sheet Recession: Cash and short-duration. Defensive equities. Avoid HY credit.

FISCAL ANALYTICS & SUSTAINABILITY

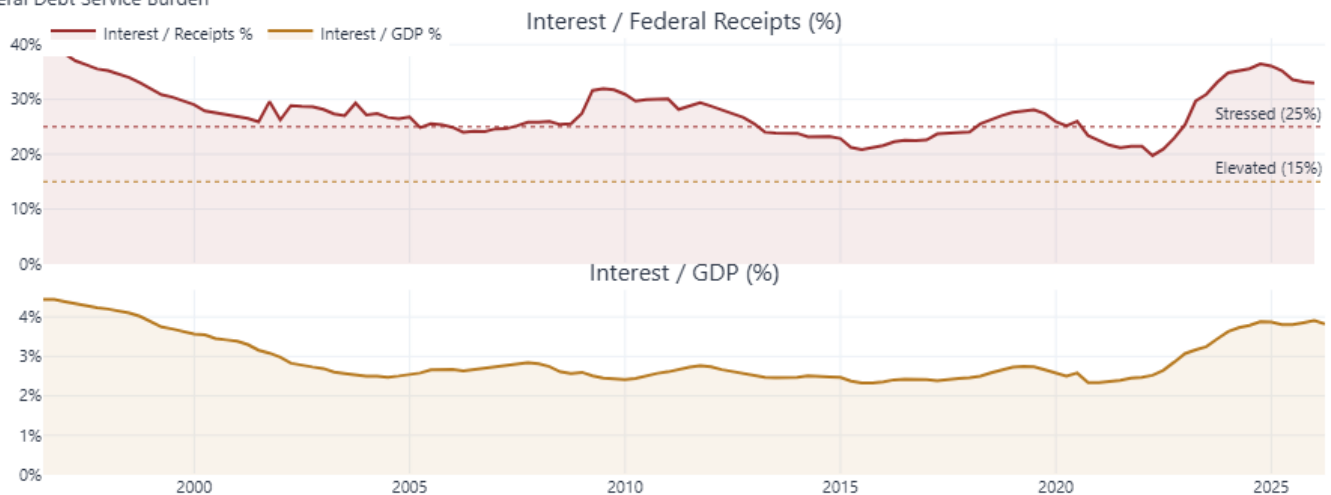
Status: Critical — Debt Service Crowding Out | Debt/GDP: 122.6% | Interest/Receipts: 32.8% | Interest/GDP: 3.83% |
Primary Balance: -5567.1% GDP | Trajectory: Broadly Stable

The Interest/Receipts ratio — not the debt/GDP level — is the most operationally meaningful sustainability metric. It answers: 'Of every tax dollar collected, how many cents go directly to paying interest before any government service is funded?' This ratio is non-discretionary: unlike defense or welfare spending, interest cannot be cut. A rising Interest/Receipts ratio crowds out all other fiscal functions.

Federal Debt / GDP — Long-Run Trajectory



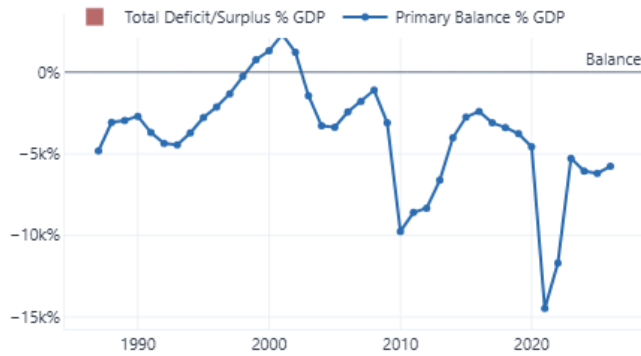
Federal Debt Service Burden



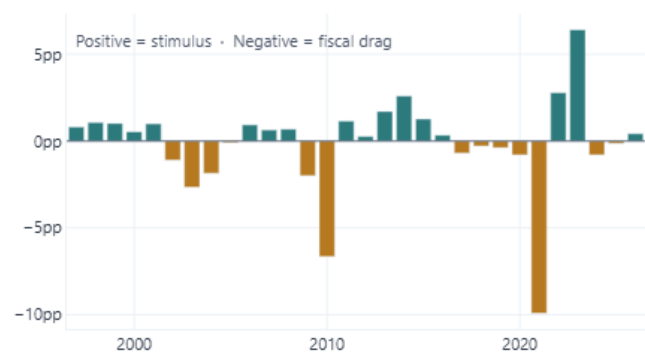
Primary Balance & Fiscal Impulse

The primary balance (total balance + interest payments) is the government's controllable fiscal position. A primary surplus of $(r - g) \times \text{debt/GDP}$ is required to stabilize the debt ratio, where r = real interest rate and g = real GDP growth. Current primary balance: -5567.1% GDP. Total deficit: -5.8% GDP (difference = interest burden of 1,219 \$B SAAR).

Total vs Primary Deficit/Surplus — % of GDP (Annual FY)



Fiscal Impulse — Annual Change in Deficit/GDP (pp)



Rollover Exposure & Trajectory

10Y yield: 4.47% vs 1.58% five years ago (change: +2.89pp). Total debt: \$38,514,009B. Estimated incremental annual rollover cost: \$+333,916B (30% annual rollover assumption). Assessment: High Rollover Cost Pressure. 5-year fiscal trajectory: debt/GDP changing at +0.15pp/yr. Interest acceleration: +155.6 \$B/yr ann.. Revenue growth: +158.3 \$B/yr ann.. The sustainability question is not whether debt/GDP is high, but whether it can be stabilized — which requires $r < g$, or a primary surplus, or both.

METHODOLOGY APPENDIX

Overview. Every indicator in this report is scored using one of four normalization methods. All methods are computed at report-generation time from the same historical data used for the charts — no pre-computed lookup tables. The methodology is intentionally transparent so that any reader can reproduce a score given the raw FRED data and the parameters documented below.

Color codes (green / yellow / red) are *descriptive*, not predictive. They indicate where the current reading sits relative to the historical distribution or a literature-calibrated threshold — not whether a recession or crisis will occur. Combinations of elevated readings across multiple independent dimensions carry more information than any single indicator.

Normalization Methods

Each indicator is assigned one of four scoring methods in the indicator registry. The method determines how the current value is translated into a risk level, and what label is displayed on the card.

Method	Used For	Green Label	Yellow Label	Red Label
Fixed Thresholds	Policy targets, literature-calibrated breakpoints	Within Range	Elevated	Stressed
Percentile (higher=worse)	Most market & macro indicators	< 75th pctile	75th–90th pctile	> 90th pctile
Percentile (lower=worse)	Labor utilization, housing starts	> 25th pctile	10th–25th pctile	< 10th pctile
Z-Score	Industrial production, structural series	Within 1σ	1– 2σ from avg	Beyond 2σ
Standardized Index	NFCI, STLFSI (pre-standardized by issuer)	Below Average	Above Average	High Stress

Calibration window. Percentile and z-score methods use a rolling historical window (20 or 30 years, documented per indicator below) ending at the most recent observation. A minimum of 24 monthly observations is required; if insufficient history is available the method falls back to fixed thresholds. The window is chosen to capture at least one full economic cycle while remaining relevant to current structural conditions.

Confidence Scoring

Each indicator card displays a confidence score (0–100). This is a composite metric that answers: *how much should I trust the current signal?* It does not affect the color coding — it qualifies it. A red signal with confidence 40 warrants more caution than a red signal with confidence 85.

Component	Weight	Description
Timeliness	25%	Days since last observation relative to expected update frequency (daily=5d, weekly=14d, monthly=60d, quarterly=120d). Stale data scores 0.
Coverage	20%	Years of history available, reaching full score at 20+ years. Insufficient history limits statistical scoring reliability.
Predictive Reliability	35%	Literature-based signal reliability assigned per indicator (0.40–1.00). Draws on Estrella & Mishkin (1998), Conference Board LEI methodology, NBER recession research, and Fed SLOOS academic literature.
Revision Risk	20%	Penalty for data subject to large initial-release revisions. Low=1.00 (daily market data, NBER), Medium=0.75 (CPI, PCE, LEI), High=0.50 (Nonfarm Payrolls, S&P; earnings).

Indicator Registry

Complete registry of all scored indicators. **Method:** normalization approach. **Win:** calibration window in years. **Rel:** predictive reliability 0–1. **Rev:** revision risk (L=Low, M=Medium, H=High). Threshold Basis summarizes the literature or statistical source for the scoring threshold.

Indicator	Method	Win	Rel	Rev	Threshold Basis
INFLATION & PRICES					
CPI	Fixed	30y	0.75	M	Fed 2% target defines green; >4.5% YoY historically required restrictive monetary policy. Yellow 2.5–4.5% = above target but not extreme.
Core CPI	Fixed	30y	0.80	M	Fed 2% target; core strips food/energy volatility. Persistent core >4% signals structural inflation requiring sustained tightening.
Core PCE	Fixed	30y	0.85	M	FOMC 2% Core PCE target; thresholds mirror Core CPI adjusted for the typical 0.3–0.5pp CPI–PCE differential.
Median CPI	Fixed	20y	0.85	L	Median CPI strips extreme price movements by using the median of component price changes. It is the best predictor of future inflation in academic literature (Stock & Watson).
Trimmed Mean PCE	Fixed	20y	0.82	L	Trimmed mean removes the top and bottom 31% of price changes by expenditure weight. Broadly comparable to median CPI in predictive power for underlying inflation.
Supercore CPI	Fixed	20y	0.78	L	Supercore strips the most volatile and persistent components. Fed Chair Powell cited supercore services (PCE equivalent) as the primary inflation measure during 2022–2023.
OER	Fixed	20y	0.65	L	OER is the largest single CPI component (~26% weight) and lags actual rents by 12–18 months. Disinflation in spot rents takes over a year to appear in CPI.
Inflation Exp (1Y)	Fixed	20y	0.65	L	Historically ~3% mean; elevated expectations risk becoming self-fulfilling. Fed tracks 'deanchoring' risk — expectations persistently above 4% signal credibility erosion.
Sticky CPI	Fixed	20y	0.80	L	Sticky prices (rents, medical, education) change infrequently, so their movement signals fundamental inflationary regime shifts rather than temporary supply shocks.
LABOR MARKET					
Unemployment	Pctile	20y	0.85	M	Percentile vs. post-1990 distribution; rate-of-change (Sahm Rule: +0.5pp above 12-month low) matters as much as level. Rising trend is the primary risk signal.
Emp/Pop Ratio	Pctile	20y	0.70	L	Percentile vs. post-2000 distribution; not distorted by participation-rate cyclicalities. Below ~59% historically aligns with labor market weakness.
LFPR	Pctile	20y	0.65	L	Percentile vs. post-2000 distribution; rolling window prevents false signals from the secular post-2000 demographic decline in participation.
Payrolls	Pctile	20y	0.85	H	Monthly additions percentile vs. post-1990 distribution; negative growth is universally adverse. Very high revision risk — first release often revised ±100K.
Init. Claims	Pctile	20y	0.85	L	Percentile vs. post-2000 distribution; leading labor indicator that rises before the unemployment rate. Weekly frequency with minimal revisions = high data confidence.
Savings Rate	Pctile	20y	0.60	M	Percentile vs. post-1990 distribution; low savings signals reduced household buffer against income shocks. Historically ~7–10% pre-2000.
Indust. Production	Z-Score	30y	0.75	M	Z-score vs. 30-year YoY distribution; consecutive negative readings define an industrial recession and have preceded every broad recession since 1970.
LEI	Fixed	20y	0.85	M	FRED USSLIND is the MoM % change of the Conference Board LEI. Negative monthly readings signal deterioration; –0.5% MoM is a sharp contractionary drop consistent with pre-recession deterioration.
CFNAI	Fixed	20y	0.80	M	CFNAI below –0.7 has reliably indicated below-trend growth; below –1.5 is consistent with recession onset (Chicago Fed guidance).
Corp. Profits	Fixed	20y	0.65	M	YoY profit growth below -5% has been consistent with recession onset; the level relative to GDP (profit margin) shows whether CAPE is inflated by above-trend margins.
Mfg New Orders	Fixed	20y	0.70	M	YoY % change in total manufacturing orders. Below 0% = contraction in order flow; sustained declines typically precede payroll contraction by 2–4 months.
Avg Weekly Hours	Pctile	20y	0.70	M	Employers cut hours before cutting headcount; this series is included in the Conference Board LEI. Percentile vs. post-1990 distribution captures the secular trend.
Temp Employment	Z-Score	20y	0.75	M	Z-score vs 20-year YoY distribution; companies reduce temp headcount 3–6 months before cutting permanent staff — a reliable early-cycle deterioration signal.
Avg Hourly Earnings	Fixed	20y	0.70	M	YoY wage growth >4% exceeds Fed comfort zone for 2% PCE target given ~2% productivity growth; >5.5% historically correlated with entrenched inflation.
ECI All Civilian	Fixed	20y	0.75	L	ECI is the Fed's preferred wage measure: less volatile than AHE. Growth above 3.5% YoY is inconsistent with 2% inflation. No revision risk (employer survey).
Job Openings	Pctile	20y	0.75	H	Percentile vs. post-2001 JOLTS history; <6M openings signals cooling labor demand. Beveridge curve: openings relative to unemployment diagnoses slack vs. mismatch.
Quits Rate	Pctile	20y	0.75	M	Percentile vs. post-2001 JOLTS history; quits rate below 2% signals workers losing confidence in job mobility — characteristic of labor market turning points.
Layoffs Rate	Pctile	20y	0.75	M	Percentile vs. post-2001 JOLTS history; layoffs rate >1.5% historically coincides with or precedes rising unemployment claims.

U-6 Rate	Pctile	20y	0.80	M	Percentile vs. post-1994 distribution (first year available). U6-U3 gap measures labor market slack beyond headline unemployment; widening gap signals hidden underemployment.
Cont. Claims	Pctile	20y	0.80	L	Percentile vs. post-2000 distribution; persistent elevated claims signal employers are not rehiring laid-off workers. Lags initial claims by ~1 week.
Productivity	Fixed	20y	0.65	H	Long-run productivity growth of ~1.5% YoY justifies non-inflationary wage gains above that level. Negative productivity signals unit labor costs rising unsustainably.
Unit Labor Costs	Fixed	20y	0.70	H	ULC > 2.5% YoY signals wages outpacing productivity — inflationary pressure that firms pass through to prices. ULC is the transmission mechanism from labor costs to goods/services prices.
Recession	Fixed	20y	1.00	L	NBER ground truth classification; no threshold scoring. Used exclusively for recession shading on charts.
MARKET VALUATION					
S&P; P/E	Pctile	30y	0.55	H	Percentile vs. post-1950 distribution; historical mean ~16x. Poor recession predictor but reliable long-run return predictor.
CAPE	Fixed	30y	0.65	M	Shiller (2000): CAPE above 25 = 'danger zone'; above 30 = peak territory historically associated with decade-long below-average real returns (1880–2025 calibration).
MARKETS					
VIX	Pctile	20y	0.75	L	VIX 20 = 75th percentile, VIX 30 = 90th percentile of post-1990 distribution — fixed thresholds coincide with well-established market-stress breakpoints.
Dollar Index	Pctile	20y	0.60	L	Percentile vs. post-2000 distribution; rapid appreciation (>8% YoY) tightens global financial conditions, pressures EM debt, and weighs on US multinationals.
INTEREST RATES					
10Y Yield	Pctile	20y	0.65	L	Percentile of post-1990 distribution; absolute level matters in context of debt/GDP — above-average rates raise refinancing burden non-linearly.
10Y Yield (Mo.)	Pctile	20y	0.65	L	Same methodology as DGS10; monthly average used for long-term trend analysis aligned with other monthly macro series.
Yield Curve	Fixed	30y	0.85	L	Estrella & Mishkin (1998): inversion precedes every US recession since 1970. Thresholds: <0% = inverted (red), 0–0.5% = flat (yellow), >0.5% = normal slope (green).
10Y-3M Spread	Fixed	30y	0.85	L	Estrella & Mishkin (1998): the 10Y-3M spread is the yield curve specification in the NY Fed recession model. Inversion (negative) has preceded every post-1970 recession.
FINANCIAL STRESS					
CRE Lending Stds	Fixed	20y	0.75	L	>40% net tightening has historically preceded CRE credit contraction by 2–4 quarters (Fed SLOOS academic literature).
Rec. Probability	Fixed	30y	0.80	L	Estrella & Mishkin (1998) NY Fed probit model: >30–40% 12-month probability has historically been a reliable recession signal. Uses 10Y–3M spread as input.
FISCAL HEALTH					
Debt/GDP	Fixed	20y	0.55	L	IMF Fiscal Monitor framework: 80% of GDP = elevated; 120%+ = high-risk trajectory for developed-market sovereign debt sustainability.
Federal Debt \$B	Fixed	20y	0.45	L	Nominal levels benchmarked to current cycle; use Debt/GDP (GFDEGDQ188S) for cross-time comparability — this series is more timely.
GDP	Fixed	20y	0.50	L	Denominator for GDP-ratio calculations; no standalone stress threshold.
Fed Deficit (\$B)	Fixed	30y	0.55	M	Deficit below -\$1.5T historically signals unsustainable fiscal trajectory absent growth acceleration. Nominal level context-dependent; use with % GDP.
Deficit % GDP	Fixed	40y	0.60	M	EU Maastricht 3% threshold is widely used. Deficit >6% in non-recessionary periods historically unsustainable without monetization.
MONEY SUPPLY					
M1	Pctile	20y	0.40	L	Percentile vs. 20-year distribution; rapid M1 expansion has led inflation with variable lags. 2020 definitional change inflated YoY readings — use with caution.
M2	Pctile	20y	0.45	L	Percentile of 20-year distribution; M2 expansion historically led inflation by 12–18 months, though the relationship weakened post-2008.
Real M2	Pctile	20y	0.55	L	Real contraction is the primary risk signal; scored by percentile of post-2000 distribution (lower percentile = worse for higher_is_bad=False).
HOUSING MARKET					
Mortgage Rate	Pctile	20y	0.65	L	Percentile vs. post-1990 distribution; affordability-adjusted threshold varies with home price levels — percentile scoring captures current rate-cycle context.
Housing Starts	Pctile	20y	0.65	M	Percentile vs. post-1990 distribution; very sensitive to mortgage rate cycles. Fixed floor of 900K represents structural contraction consistent with prior recessions.
Home Prices	Pctile	20y	0.60	L	Percentile vs. post-2000 YoY distribution; >8% YoY signals affordability stress and speculative dynamics consistent with past boom-bust episodes.
Building Permits	Pctile	20y	0.70	M	Percentile vs. post-1990 YoY distribution; leading indicator for residential construction and related employment. Sustained YoY contraction precedes housing-sector recessions.
LIQUIDITY & FUNDING STRESS					

FSI	Std. Index	20y	0.80	L	Index centered at 0 (mean conditions since 1993); +1.5 ≈ 90th percentile of tightening episodes, representing significant systemic stress.
3M CP Rate	Fixed	20y	0.65	L	Supporting series for CP-Treasury spread; absolute level driven by policy rate + credit premium — spread vs. Treasuries is the actionable signal.
3M Treasury	Fixed	20y	0.65	L	Short-term risk-free rate; levels reflect Fed policy stance. Used primarily as CP-Treasury spread denominator — standalone level is policy-cycle dependent.
SOFR	Fixed	20y	0.60	L	No standalone stress thresholds; LIBOR replacement tracks Fed Funds rate. Dislocations vs. FEDFUNDS signal repo funding stress.
Reverse Repo	Fixed	20y	0.65	L	Calibrated against the 2019 reserve-scarcity episode where repo rate spikes followed RRP near-depletion. Single-episode calibration: RRP drawdown can also reflect reserve redistribution or T-bill issuance absorbing MMF demand — not a universal rule.
NFCI	Std. Index	20y	0.80	L	Chicago Fed constructs the index with mean 0 (1971 base); +0.7 ≈ 90th percentile of tightening episodes. Combines 105 indicators across risk, credit, and leverage subindices.
CREDIT MARKET FUNCTIONALITY					
HY Spread	Pctile	20y	0.80	L	Percentile vs. post-1997 ICE BofA dataset; ~4% ≈ 75th percentile, ~7% ≈ 90th percentile of full-cycle distribution.
Baa Spread	Pctile	20y	0.75	L	Historical average ~2.0%; spikes to 3.5%+ during financial stress (2002, 2009, 2020). Available monthly from 1986.
CC Delinquency	Pctile	20y	0.70	L	Percentile vs. post-1991 distribution; consumer credit health is a lagging indicator — rises after income stress is established. Peaked at ~6.8% in 2009.
IG Spread	Pctile	20y	0.75	L	Percentile vs. post-1997 ICE BofA dataset; widening above 2.5% signals credit tightening reaching investment-grade issuers — a systemic stress indicator.
NFCI Credit	Std. Index	20y	0.75	L	Credit-specific subindex of NFCI; same mean-0 construction. Isolates credit channel stress from leverage and risk components.
Lending Stds	Fixed	20y	0.80	L	Bernanke & Lown (1991): >40% net tightening historically precedes C&I; credit contraction by 2–4 quarters. SLOOS survey with quarterly frequency.
Delinquency (All)	Pctile	25y	0.75	M	Percentile vs. post-1990 history; >2.5% has preceded systemic credit events. Rapid 6-month increase matters as much as level.
CRE Delinquency	Pctile	25y	0.80	M	CRE concentrated at regional banks; >2.5% historically preceded regional banking stress episodes (S&L; crisis, GFC). Office/retail vacancy surge is a 2024-onwards risk.
Mortgage Delinquency	Pctile	25y	0.75	M	Percentile vs. post-1991 history. Mortgage delinquencies >3% (GFC reached 11%) typically require systemic government response. Rising from low base warrants monitoring.
Bank Deposits	Fixed	20y	0.70	L	Deposit outflows are a direct banking stress indicator. YoY contraction >3% signals systemic deposit flight risk. SVB 2023 highlighted speed of modern bank runs.
POLICY CONSTRAINTS					
Fed Assets	Fixed	20y	0.50	L	No stress thresholds — balance sheet size has no universal threshold; displayed as % of GDP for comparability across QE/QT cycles.
Fed Funds	Fixed	20y	0.70	L	No standalone stress thresholds; the real rate (FEDFUNDS – Core CPI) is the relevant policy-stance metric, not the nominal level.
5Y BEI	Fixed	20y	0.75	L	2.5% = Fed target + 50bp buffer (green/yellow boundary); 3.2%+ = unanchoring territory constraining the Fed's ability to ease (Bernanke/Yellen framework).
10Y BEI	Fixed	20y	0.75	L	Long-run expectations above 3% would represent significant unanchoring not seen since the 1980s — severely constraining Fed's ability to respond to economic weakness.
Interest Pmts	Fixed	20y	0.55	L	Supporting data for interest/receipts ratio; no standalone threshold scoring.
Fed Receipts	Fixed	20y	0.50	L	Supporting data for interest/receipts ratio denominator; no standalone threshold scoring.
GLOBAL MACRO & FX					
Brent Crude	Pctile	20y	0.65	L	Percentile vs. 20Y history. >50% YoY surge historically transmits to headline CPI within 3–6 months. Rapid declines signal demand destruction.
PPI Commodities	Pctile	30y	0.70	M	Percentile vs. 30Y history. PPI leads CPI by ~2-3 months. >15% YoY sustained = significant inflation pass-through risk.
EUR/USD	Pctile	20y	0.55	L	Rapid EUR/USD decline (>10% YoY) = sharp dollar strengthening = global financial tightening impulse.
JPY/USD	Pctile	20y	0.50	L	Rising JPY/USD = yen depreciation = BOJ/Fed policy divergence. >15% YoY yen weakening signals BoJ yield curve control stress.
CNY/USD	Pctile	20y	0.55	L	Yuan depreciation >8% YoY historically triggers global EM contagion and competitive devaluation fears.
ECB Rate Proxy	Fixed	25y	0.60	L	ECB's own 2% inflation target implies neutral rates near 2–2.5%. Above 4% = historically restrictive for the eurozone.
BOJ Rate	Fixed	25y	0.60	L	BOJ has operated near zero since 1999; any rate normalization above 1% signals a historic regime shift with global carry-trade implications.
STRUCTURAL & LONG-RUN					
Real GDP	Pctile	30y	0.85	H	Positive real GDP growth above 2% (trend) = green; below trend but positive = yellow; negative = red (recession signal).
Potential GDP	Fixed	30y	0.70	M	CBO's estimate of the economy's non-inflationary production capacity. Used to compute the output gap; no standalone threshold.
Working Age Pop.	Fixed	20y	0.65	L	Growth below 0.2% YoY = significant demographic headwind for labor supply and potential GDP growth.

10Y TIPS Yield	Fixed	20y	0.80	L	Real rates above 2.5% are historically restrictive — they compress equity multiples, slow housing, and pressure leveraged borrowers.
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Limitations & Caveats

Survivorship in calibration windows	The calibration window includes only the period for which data is available. Indicators with short histories (e.g., SOFR since 2018) cannot be reliably percentile-scored and fall back to fixed thresholds.
Regime shifts	A 20-year rolling window may include structural breaks (e.g., post-2008 zero-rate environment, post-2020 M1 definitional change). Percentile scores calibrated against a structurally different regime may mis-classify current readings.
Revision risk	Many indicators (Nonfarm Payrolls, GDP, PCE) are substantially revised after initial release. The confidence score penalizes high-revision series, but the scored value is always the latest available — which may itself be revised.
Composite vs. individual	Crisis dimension scores use worst-of aggregation. This can be overly conservative: a single elevated indicator can lift an entire dimension even when the other components are benign.
No prediction	All scores are backward-looking relative to the calibration window. A reading at the 95th percentile of history tells you it is historically unusual — it does not predict that a crisis will follow.

Key References

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Chicago Fed NFCI Methodology	Brave & Butters (2011), Monitoring Financial Stability: A Financial Conditions Index Approach. Economic Perspectives Q1 2011. Basis for NFCI index interpretation.

INDICATOR GLOSSARY

Plain-English definitions for all acronyms and metrics used in this report. Click any indicator name in the Summary section above to jump to detailed charts.

CPI	Consumer Price Index — average price change for a fixed basket of urban household goods	Delinquency Rate	Share of loans past due (all loans, CRE, or mortgages); rising rate signals deteriorating credit quality
Core CPI	CPI excluding food & energy; strips volatile components to show underlying trend	Bank Deposits	Total deposits at commercial banks (DPSACBM027SBOG); sharp decline signals deposit flight or tightening liquidity
Core PCE	PCE Price Index ex food & energy — the FOMC's explicit 2% inflation target; published by BEA monthly	Charge-off Rate	Loans written off as unrecoverable as a % of total loans; lags delinquencies by 1–3 quarters
CPI vs PCE	PCE typically runs 0.3–0.5% below CPI due to different weights and substitution methodology	VIX	CBOE Volatility Index — market's 30-day implied volatility; the 'fear gauge'; > 30 signals elevated stress
Supercore CPI	CPI for services excluding food, energy, and shelter — closely tracks wage growth; the Fed's preferred cyclical inflation gauge	CAPE / Shiller P/E	S&P; 500 price divided by 10-year inflation-adjusted average earnings; > 30 historically precedes lower future returns
Shelter CPI	CPI component for housing costs (~35% of CPI); dominated by OER and lags real-time rents by 12–18 months	Trailing P/E	Stock price divided by last 12 months of reported earnings
Median CPI	Cleveland Fed measure using the median price change across all CPI components; filters outliers better than core CPI	10Y Treasury Yield	Benchmark long-term government rate; affects all asset prices via the risk-free rate
Trimmed Mean PCE	Dallas Fed measure that removes top and bottom price-change outliers each month; smoothest real-time inflation signal	Fed Funds Rate	Fed's overnight policy rate set by the FOMC; the primary monetary policy tool
Sticky CPI	Atlanta Fed CPI for goods/services with infrequent price changes (e.g., rents, insurance); best predictor of persistent inflation	Real FF Rate	Fed Funds Rate minus Core CPI YoY — the inflation-adjusted policy stance; negative = accommodative
Flexible CPI	CPI for goods/services with frequent price resets (e.g., gasoline, airfare); leads overall CPI by several months	QE / QT	Quantitative Easing (buying assets to inject reserves) / Tightening (shrinking the Fed balance sheet)
OER	Owners' Equivalent Rent — imputed monthly rent a homeowner would pay; ~26% of CPI weight with a structural 12–18 month lag	WALCL	Fed balance sheet total assets; expanded via QE, shrunk via QT
BEI / Breakeven	Market-implied inflation expectations from the gap between TIPS and nominal yields	SAAR	Seasonally Adjusted Annual Rate — removes seasonal patterns, expressed at annual pace
T5YIE / T10YIE	5- and 10-year breakeven inflation rates from TIPS pricing	Debt / GDP	Federal debt as a percentage of gross domestic product; above 100% limits fiscal flexibility
TIPS	Treasury bonds whose principal adjusts with CPI; used to extract inflation expectations	Interest / Receipts	Federal interest payments as a share of total government revenues; rising ratio crowds out discretionary spending
Michigan Survey	University of Michigan 1-year household inflation expectations survey; sensitive to gas prices and political sentiment	Yield Curve (10Y-2Y)	Spread between 10-year and 2-year Treasury yields; inversion has preceded every US recession since 1970
Deanchoring	Loss of public confidence that inflation will return to target; self-fulfilling through wage and price setting behavior	Inverted Curve	When short rates exceed long rates (spread < 0) — markets pricing near-term risk higher than long-run growth

Cyclical Inflation	Demand-driven price pressure — tied to labor market tightness, services spending, wage growth; sensitive to monetary policy	NY Fed Rec. Prob.	Model-based 12-month recession probability using yield curve slope; above 30% has historically been a reliable signal
Structural Inflation	Supply-side or cost-push inflation — energy shocks, supply chain disruptions, demographic shifts; slower to respond to rate hikes	Housing Starts	New residential units started monthly (SAAR); a leading indicator for construction employment and materials
Inflation Regime	Classification of current inflation environment: At Target (<2.5%), Disinflation, Elevated, or Entrenched (sticky >4%)	Case-Shiller HPI	S&P;/Case-Shiller repeat-sales home price index; published with ~2-month lag
M1	Narrow money — physical currency plus checking-account deposits	Mortgage Rate (30Y)	Freddie Mac Primary Mortgage Market Survey; directly affects housing affordability and purchase volume
M2	Broad money — M1 plus savings accounts, money-market funds, and small CDs	Dollar Index (DTWEX)	Trade-weighted U.S. dollar index vs. 26 currencies; rapid appreciation tightens global financial conditions
Real M2	M2 adjusted for inflation; sustained contraction signals tightening conditions	Brent Crude	European benchmark for global oil pricing; closely tracked because it sets the reference price for ~2/3 of world oil contracts
U-3 (Unemployment)	Official unemployment rate — share of the labor force actively seeking but unable to find work	PPI Commodities	Producer Price Index for all commodities; a broad upstream inflation gauge that leads CPI by 2–3 months
U-6 (Underemployment)	Broad labor underutilization: unemployed + marginally attached workers + part-time for economic reasons	Gold Price	Safe-haven and inflation hedge; rising gold reflects dollar stress, elevated inflation fear, or geopolitical risk premium
U6–U3 Gap	Difference between U-6 and U-3; widening gap signals rising hidden slack — workers underemployed, not counted as unemployed	DXY / Broad USD	Trade-weighted U.S. dollar index; strong dollar tightens global financial conditions for dollar-denominated debt holders
LFPR	Labor Force Participation Rate — working-age adults in the labor force (%)	EUR/USD	Most-traded FX pair globally; reflects Fed-ECB policy divergence and Eurozone growth/risk premium vs. the U.S.
Emp/Pop Ratio	Share of all working-age adults employed; unaffected by participation shifts	JPY/USD	Yen per dollar; rising JPY/USD = weaker yen = BOJ/Fed divergence. Yen is the world's primary carry-trade funding currency
Sahm Rule	Recession indicator triggered when the 3-month avg unemployment rate rises 0.5pp above its 12-month low	CNY/USD	Yuan per dollar; rapid depreciation signals China growth stress or capital outflows with global EM contagion risk
Nonfarm Payrolls	Net monthly job additions across all non-agricultural sectors	ECB Rate	European Central Bank deposit facility rate; the benchmark monetary policy instrument for the 20-nation Euro Area
AHE	Average Hourly Earnings — monthly wage growth measure; YoY above 4% in a tight market signals wage-price pressure	BOJ Rate	Bank of Japan overnight call rate; BOJ normalization from near-zero is a major global tail risk via yen carry unwind
ECI	Employment Cost Index — quarterly measure of total compensation costs including benefits; less volatile than AHE	CB Divergence	Gaps between major central bank policy rates; drives FX, capital flows, and global financial conditions asymmetries
Initial Claims	Weekly new unemployment insurance filings (ICSA); a timely leading indicator of layoff activity	Macro Regime	Classification of the current macro environment based on growth, inflation, financial conditions, and credit dimensions
Continued Claims	Number of people currently receiving unemployment benefits (CCSA); reflects difficulty finding new jobs	Goldilocks	Above-trend growth + contained inflation + neutral FCI — the optimal backdrop for broad risk appetite

Absorption Ratio	Ratio of continued claims to initial claims; rising ratio means fired workers are taking longer to find new jobs	Reflation	Above-trend growth + rising inflation — commodity and cyclical outperformance; central banks may be behind the curve
Temp Employment	Temporary help services payrolls (TEMPHELPS); often turns negative 3–6 months before broader job losses	Stagflation	Below-trend growth + elevated inflation — worst policy trade-off; rate cuts risk entrenching inflation
JOLTS	Job Openings and Labor Turnover Survey — monthly BLS survey of open positions, hires, quits, and layoffs	Disinflation	Softening growth + moderating inflation — rate cuts become viable; duration assets typically outperform
Job Openings	Total unfilled positions (JTSJOL); > 10M openings historically signals very tight labor conditions	Liquidity Boom	Easy financial conditions + credit expansion + asset price inflation — watch for late-cycle excess
Quits Rate	Share of employed workers voluntarily quitting (JTSQUR); high rate = workers confident in finding better jobs	Tightening Cycle	Rising rates + tightening FCI — overtightening risk (inversion, spread widening, claims acceleration)
Layoffs Rate	Share of employed workers involuntarily separated (JTSLDR); rising rate = employer-driven labor market deterioration	Balance-Sheet Recession	Credit collapse + deflation risk — the defining challenge is debt deleveraging, not monetary policy adjustment
Beveridge Curve	Relationship between job openings and unemployment; shift outward = matching inefficiency or structural mismatch	Dimension Score	Normalized 0-to-1 risk score for each macro dimension: 0 = healthy, 1 = severely stressed
V/U Ratio	Vacancies-to-Unemployed ratio; above 1.0 means more jobs than job seekers — historically rare and inflationary	Output Gap	$(\text{Real GDP} - \text{Potential GDP}) / \text{Potential GDP} \times 100\%$; positive = overheating, negative = economic slack
Nonfarm Productivity	Output per hour worked in the nonfarm business sector (OPHNFB); rising productivity allows wages to grow without inflation	Potential GDP	CBO's estimate of the economy's non-inflationary production capacity — sets the benchmark for the output gap
ULC	Unit Labor Costs — labor cost per unit of output (ULCNFB); sustained rise above 3% YoY signals inflation pressure	r* (Neutral Rate)	The real interest rate consistent with full employment and stable inflation. Unobservable; estimated with state-space models (e.g., Holston-Laubach-Williams). Often proxied by the 10Y TIPS yield, though TIPS also embeds a real term premium (~0.5–1.5%) that causes it to overstate r*, especially during tightening cycles.
Real Wage Growth	Nominal wage growth minus CPI; negative real wages erode purchasing power even when nominal wages rise	Real FF Rate	Nominal Fed Funds Rate minus core CPI YoY — policy stance indicator: above the 10Y real rate = restrictive; below = accommodative
LDI	Labor Deterioration Index — composite 0–100 score weighting unemployment, claims, JOLTS, and payrolls; >60 = significant stress	10Y TIPS Yield	Yield on 10Y inflation-protected Treasuries. A market-observable long-run real rate benchmark, but not equivalent to r*: the TIPS yield also embeds a real term premium and a TIPS liquidity discount, both absent from the neutral rate concept.
LEI	Conference Board Leading Economic Index — composite of 10 forward-looking indicators; sustained decline signals recession	Working Age Pop.	Population aged 15–64 (FRED: LFWA64TTUSM647S); slow-moving driver of labor supply and long-run potential GDP
BCI	Business Cycle Indicator — normalized composite of leading indicators scaled 0–100; <40 = contraction territory	Globalization Index	Proxy from USD trend: dollar appreciation signals deglobalization; commodity supercycles track globalization phases
ISM PMI	ISM Manufacturing Purchasing Managers' Index; above 50 = expansion, below 50 = contraction	Interest/Receipts	Federal interest payments as % of total tax receipts — the fraction of each tax dollar consumed by debt service
ISM New Orders	ISM sub-index of new manufacturing orders; the most forward-looking component of the PMI	Primary Balance	Budget balance excluding interest payments; the government's controllable fiscal position and key sustainability metric

Building Permits	Monthly new residential building permits (SAAR); a leading indicator for housing activity 3–6 months ahead	Primary Deficit	Primary balance below zero — government spending (ex-interest) exceeds revenues; the most concerning fiscal signal
NBER Recession	Official U.S. recession dates determined by the NBER Business Cycle Dating Committee; typically declared months after onset	Fiscal Impulse	Year-over-year change in the deficit/GDP ratio; positive = fiscal stimulus (expanding deficit), negative = fiscal drag
Diffusion Index	Share of indicators improving minus share deteriorating; above 50 = net expansion, below 50 = net contraction	r > g Risk	When real interest rates exceed real GDP growth, debt/GDP rises without a primary surplus — the debt spiral condition
FCI	Financial Conditions Index — composite of rates, spreads, equity prices, and volatility into a single tightness score	Rollover Risk	Risk that maturing debt must be refinanced at materially higher rates, increasing the interest burden mechanically
NFCI	National Financial Conditions Index — Chicago Fed weekly gauge across 105 variables; negative = easy, positive = tight	Debt Stabilization	Condition where debt/GDP stops rising; requires: primary surplus $\geq (r - g) \times \text{debt/GDP}$, or sufficiently high growth
STLFSI	St. Louis Fed Financial Stress Index — weekly composite; values above 0 indicate above-average financial stress	Recession Probability	Model-generated probability (0–100%) of recession onset within a specified horizon (6M / 12M / 24M)
SOFR	Secured Overnight Financing Rate — overnight repo benchmark; LIBOR replacement since 2023	Logistic Regression	Statistical model that maps macro indicator values to a probability between 0% and 100% via a sigmoid function
Reverse Repo (RRP)	Fed facility where counterparties park cash overnight; high usage = ample reserves parked at the Fed; drawdown reflects reserve redistribution, T-bill issuance absorbing MMF demand, or genuine liquidity reduction — context required	Feature Contribution	Each indicator's additive contribution to the total recession probability score; identifies the dominant risk drivers
NFCI Credit	NFCI subindex isolating credit conditions specifically; most sensitive component to bank lending tightness	Confidence Band	Bootstrap-estimated uncertainty range around the central probability; wider bands = less historical data or regime change
CP-Tsy Spread	Commercial paper rate minus T-bill rate — short-term funding stress indicator; spikes during financial crises	OOS Backtest	Out-of-sample backtest — model trained on data before a given date, tested on data it never saw during training
HY Spread (OAS)	Extra yield demanded above Treasuries for below-investment-grade bonds; > 600bps signals distress	Hit Rate	Share of recessions correctly flagged by the model above a chosen probability threshold
IG Spread (OAS)	Extra yield demanded above Treasuries for investment-grade corporate bonds	False Positive Rate	Share of non-recession periods incorrectly flagged as recession risk; the trade-off against hit rate
OAS	Option-Adjusted Spread — yield spread net of embedded call/put option value	Z-Score	Standard deviations from the historical mean; $\pm 2\sigma$ marks the outer 5% of a normal distribution
SLOOS	Senior Loan Officer Opinion Survey — quarterly Fed survey on bank lending standards and demand	Percentile	Rank of the current reading vs. historical distribution; 90th percentile = higher than 90% of all past readings
Lending Standards	Net % of banks tightening loan conditions; tightening > 40% historically signals credit contraction ahead	YoY / MoM	Year-over-Year / Month-over-Month percent change — annualized and monthly rates of change
CRE	Commercial Real Estate — offices, retail, apartments, and industrial property	Rolling Window	Historical lookback period for computing statistics; e.g., 20-year window updates each month as new data arrives
CRE Lending Stds	Net % banks tightening CRE loan standards (SLOOS); > 40% signals credit contraction ahead	Revision Risk	Magnitude of typical revisions to initial data releases; high for payrolls and GDP, low for daily market prices
CC Delinquency	Share of credit card loans past due — broad consumer credit health signal		